

Opaque Trading, Disclosure, and Asset Prices: Implications for Hedge Fund Regulation

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We investigate the effect of ambiguity about hedge fund investment strategies on asset prices and aggregate welfare. We model some traders (mutual funds) as facing ambiguity about the equilibrium trading strategies of other traders (hedge funds). This ambiguity limits the ability of mutual funds to infer information from prices and has negative effects on market outcomes. We use this analysis to investigate the implications of regulations that affect disclosure requirements of hedge funds or the cost of operating a hedge fund. Our analysis demonstrates how regulations affect asset prices and welfare through their influence on opaque trading. (*JEL* G14, G12, G11, D82)

Hedge funds control nearly two trillion dollars of investable funds. Although this amount is small relative to the total assets under management by mutual funds (Vanguard alone has about \$1.6 trillion), it is large enough to move markets, and in the aftermath of the financial crises, big enough to concern regulators. Perhaps more important than their size, however, is the information that hedge funds have about their own trading strategies and about future asset values. Unlike mutual funds, which are required by the U.S. Securities and Exchange Commission (SEC) to disclose publicly their holdings and positions on a regular basis, hedge funds face much more limited disclosure requirements.

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Current rules require hedge funds with \$1 billion or more under management to file quarterly reports with regulators describing their trading and portfolio positions, but these reports are not disclosed to the public. There is now ongoing debate as to whether hedge funds should face the same public disclosure requirements as mutual funds.¹

Hedge funds offer a number of arguments for their need for opacity. Unlike mutual funds, which are offered to the public, hedge funds are restricted to private investors. Disclosure rules for mutual funds facilitate monitoring by both current and future investors, but such monitoring for hedge funds can be accomplished by (less costly) internal disclosures. Moreover, forcing public disclosure by hedge funds could allow others to infer their trading strategies and information, inducing mimicking trade, which could erode the profitability of strategies.² In addition, the large scale of hedge fund positions means that any position adjustments impose liquidity demands on the market. Knowledge of fund positions could allow others to exploit this need for liquidity, thereby also reducing the profitability of hedge fund trading. For a variety of reasons, therefore, it is in the best interest of hedge funds to keep their trading secret and in the best interest of others in the market to discover as much as possible about their trading in order to profit from it.³

In this research, we investigate the implications of hedge fund regulation for asset prices and welfare. Our particular focus is on rules requiring hedge funds to disclose their trading, and other regulations that change hedge funds' cost of doing business.⁴ No useful conclusions can be reached about the impact of hedge fund regulation until we understand the implications of opacity of trading strategies for both individual investors and the market.

To address this issue, we construct a rational expectations equilibrium model in which some traders (hedge funds) have private information, and elements of trading strategies employed by these traders are not known to other traders (mutual funds). We model this uncertainty by assuming that the trading strategies of hedge funds are viewed as being ambiguous by mutual funds. Our focus on ambiguity captures what we believe is an important aspect of hedge

¹ As discussed in Aragon and Nanda (2012), Title IV of Dodd Frank directs the SEC to impose regulatory requirements on hedge funds as it deems in the public interest or for the assessment of systemic risk. How, exactly, such requirements are to be implemented remains contentious. For example, in meeting new requirements to register as investment advisors, 150 prominent hedge funds did so by masking the true names of their funds, effectively shielding information from outside investors. See "Who is that masked hedge fund," *Wall Street Journal*, June 22, 2012.

² See Reca, Sias, and Turtle (2012), who analyze the impacts of herding in the hedge fund industry.

³ Aragon, Hertz, and Shi (2012) find that confidential hedge fund positions earn significant, positive abnormal returns and that this profitability is an important determinant of the decision by hedge funds to maintain confidentiality.

⁴ Of course, regulations that impact hedge funds may have other implications for the economy that are outside of our analysis. One interesting possibility would be to consider regulations that address the systemic risk created by interdependence among hedge fund strategies. See Boyson, Stahel, and Stulz (2010). Analyzing this impact would require modeling systemic risk, which goes beyond the scope of this paper.

fund opacity: it is not just that hedge funds may have information that is not available to other traders, but rather that other traders are not even sure what hedge funds are doing. This uncertainty about strategies introduces a variety of complications not captured by simple models of asymmetric information, and it affects the incidence and efficacy of hedge fund regulation.

Hedge funds in our analysis have access to private investment opportunities not available to mutual funds, a construct also found in Wang (1994). These extra investment opportunities could be direct investments in companies, options trading, short-selling capabilities, and asset classes in which mutual funds are not permitted to invest.⁵ The correlations between returns on these investment and returns on the stock market are also the source of ambiguity about the hedge funds' trading strategies. This correlation allows hedge funds to diversify in ways unavailable to mutual funds, and it results in hedge funds' effective risk tolerance being higher than that of mutual funds. The ambiguity that mutual funds face about this effective risk tolerance interferes with their ability to infer the information held by hedge funds, and is one source of the attractiveness of being a hedge fund.

After describing the equilibrium with a fixed fraction of hedge funds, we find the equilibrium fraction of hedge funds by allowing institutions to choose between operating as a hedge fund or a mutual fund.⁶ This allows us to investigate the impact of disclosure regulations affecting ambiguity about hedge fund strategies and regulations changing the cost of operating a hedge fund. In our model, these regulatory changes affect the equilibrium fraction of hedge funds, and hence they may have unforeseen effects on the economy. For example, we find that increasing the differential cost of operating a hedge fund will decrease the fraction of hedge funds, increase the equity premium, and decrease welfare. Thus, regulatory policies designed to limit the number of hedge funds simply by increasing their costs of doing business seem ill advised.

Regulatory policies designed to reduce hedge fund ambiguity, such as enhanced disclosure rules, have more complex effects. Not surprisingly, disclosure reduces the ambiguity (and profitability) of hedge funds strategies, and so reduces the number of hedge funds in equilibrium. The effects on the equity premium and welfare are less clear. Reducing ambiguity makes mutual funds better able to use prices to infer the information held by hedge funds, and this tends to reduce the equity premium; but, reducing ambiguity also reduces

⁵ Note that these extra investment opportunities make hedge funds socially valuable. Without these opportunities there would be no reason in our model for the existence of hedge funds. It would not be informative to study optimal hedge fund regulation in an environment in which they should not exist, as the optimal regulation would simply prohibit them. The social value arising from the extra investment opportunities available to hedge funds is partially offset by the ambiguity that hedge funds introduce, and it is the balance of these two effects that we study in our efficiency and welfare analysis.

⁶ We do not consider competition between hedge funds and mutual funds arising from competition for managers or alternative compensations structures. For analysis of these issues, see Deuskar, Pollet, Wang, and Zheng (2011).

the number of hedge funds, and this tends to raise the equity premium. Similarly, the effect on welfare is ambiguous. Our analysis isolates these two offsetting effects, but it remains an empirical question whether greater disclosure has the effects that regulators desire.

Two key aspects of our modeling strategy are the assumptions that there is asymmetric information and that mutual funds are ambiguity averse about hedge fund strategies. We believe that it is natural to assume that hedge funds develop information that is not available to mutual funds, and so we assume that they have private information that they use in their trading. Mutual funds, of course, infer as much about this information as they can from market statistics. But the ambiguity that they face about hedge fund strategies interferes with this inference. This ambiguity is crucial for our results. Many of our results (in particular, those related to disclosure policies aiming to improve transparency) vanish or are reversed if there is asymmetric information but no ambiguity aversion. We show this by extending our analysis to the more general case of smooth ambiguity aversion and showing that our results hold only if there is sufficient ambiguity aversion (see Section 5). We believe that ambiguity about hedge fund strategies is natural, so we view its role in delivering our results as a positive attribute of our model.

Note, however, that ambiguity about hedge fund strategies by itself, without asymmetric information, is not sufficient for our results. Unless hedge funds have private information that is directly relevant to payoffs of the assets in which mutual funds invest, ambiguity about hedge fund strategies is irrelevant. This occurs because in a market equilibrium setting no trader actually cares directly about any other trader's demand.⁷ It is only because ambiguity interferes with the ability of mutual funds to infer the private information held by hedge funds that ambiguity matters (see Section 4). Thus our analysis highlights both the unique and interdependent roles played by asymmetric information and ambiguity in affecting equilibrium outcomes.

We also extend our model to show that our results are robust to a variety of other, more general alternate specifications. For example, in Appendix A, we allow mutual funds to be ambiguous about multiple dimensions of the aggregate trading of hedge funds, and show that our results continue to hold.

Our paper is most closely related to recent papers studying the process of information transmission through prices in the presence of ambiguity (Caskey 2009; Condie and Ganguli 2011; Mele and Sangiorgi 2011; Ozsoylev and Werner 2011).⁸ Caskey (2009) uses Klibanoff, Marinacci, and Mukerji's (2005) smooth ambiguity aversion to demonstrate that ambiguity-averse investors

⁷ In a competitive equilibrium, every trader assumes (correctly) that they can buy or sell as much of the asset as they want at the prevailing market price. Unless this price also conveys information, that is all traders need to know. What trading strategies others used in deciding on their demands is irrelevant.

⁸ Ambiguity aversion has been extensively used to explain many interesting phenomena in financial markets, including, *inter alia*, portfolio choice and non-market participation (Dow and Werlang 1992; Uppal and Wang 2003; Cao, Wang, and Zhang 2005; Garlappi, Uppal, and Wang 2007; Easley and O'Hara 2009;

may prefer aggregate information even when the aggregate signal is not a sufficient statistic for its components, thereby causing prices to underreact to public information. Condie and Ganguli (2011) demonstrate the existence and robustness of partially revealing rational expectations equilibria in economies with ambiguity-averse preferences. Mele and Sangiorgi (2011) and Ozsoylev and Werner (2011) analyze noisy rational expectations equilibria in economies with Gilboa and Schmeidler's (1989) max-min ambiguity aversion preferences and study implications of ambiguity for volume, liquidity, and price volatility.

Our paper is different from and thus complements these papers in two ways. First, the source of ambiguity is different. In these studies, investors are ambiguous about asset payoffs, or exogenous and public fundamental signals, such as earnings reports and analyst forecasts. In contrast, in our economy, investors are not ambiguous about fundamentals, but instead are ambiguous about the trading strategies of some sophisticated traders, whose trading in equilibrium causes prices to become an endogenous ambiguous signal regarding the asset payoff. Second, these studies, except Condie and Ganguli (2011), use noise trading to prevent fully revealing prices. The unmodeled noise trading renders a complete welfare analysis impossible. Given that one of our main focuses is on the implications of regulation, welfare analysis is particularly important. In this sense, our model without exogenous noise trading is more useful for the questions we ask.

In regard to the general issue of welfare effects when some traders are uncertain about the trading of other traders, our paper is also related to Stein (1987), who shows that introducing new speculators into the market can negatively affect market efficiency and reduce the aggregate welfare. But the approach and mechanism in Stein's (1987) model are very different from ours. Stein (1987) employs a traditional risk-aversion approach and assumes that the noisy signal of new speculators is not observable to those traders who are already in the market. Under some conditions, this feature can generate price destabilization, which in turn leads to welfare reduction. In contrast, our model employs an ambiguity approach to capture mutual funds' uncertainty about hedge fund trading; the uncertainty is not about hedge funds' private information, but rather about how they trade on this information. The ambiguity created in this way has no effect on price volatility, and its negative welfare effect occurs primarily through an equity premium channel.

Finally, our paper is related to the large, and growing, literature on hedge funds and their impact on the economy. An excellent survey of work in this area is Stulz (2007). More recent work has focused on a wide range of issues, such as

Epstein and Schneider 2010; Cao, Han, Hirshleifer, and Zhang 2011; Illeditsch 2011), the equity premium puzzle (Epstein and Wang 1994; Uti 2011), learning (Epstein and Schneider 2007, 2008; Ju and Miao 2012), liquidity (Caballero and Krishnamurthy 2008; Routledge and Zin 2009), market selection and survival (Condie 2008), and microstructure (Easley and O'Hara 2010a,b). Several recent studies empirically investigate the impact of ambiguity with real market data (e.g., Leippold, Trojani, and Vanini 2008; Anderson, Ghysels, and Juergens 2009).

systemic risk concerns posed by hedge funds (Boyson, Stahel, and Stulz 2010), hedge fund herding (Reca, Sias, and Turtle 2012), mutual fund and hedge fund interaction (Deuskar, Pollet, Wang, and Zheng 2011), hedge fund managerial incentives (Aragon and Nanda 2012), and disclosure (Aragon, Hertz, and Shi 2012). Our work adds to this literature by looking at the effects of ambiguity about hedge fund strategies on market performance and regulation.

1. The Model

1.1 Environment

In the economy we analyze, two assets are traded: a risk-free asset, the bond, which has a constant value of 1; and, a risky asset, the stock, which has a price of $\bar{p}$ per unit and an uncertain future value $\tilde{v}$. We assume that

$$\tilde{v} = \bar{v} + \tilde{\theta}_T + \tilde{\theta}_O + \tilde{\varepsilon}, \quad (1)$$

where $\bar{v} > 0$, $\tilde{\theta}_T \sim N(0, \sigma_{\theta T}^2)$ with $\sigma_{\theta T} > 0$, $\tilde{\theta}_O \sim N(0, \sigma_{\theta O}^2)$ with $\sigma_{\theta O} > 0$, $\tilde{\varepsilon} \sim N(0, \sigma_{\varepsilon}^2)$ with $\sigma_{\varepsilon} > 0$, and $\tilde{\theta}_T$, $\tilde{\theta}_O$ and $\tilde{\varepsilon}$ are mutually independent. One can view this risky asset as a proxy for the stock market.

There is a $[0, 1]$ continuum of rational traders who are initially identical. Each trader is endowed with one share of the stock and zero units of the bond. These traders have constant absolute risk aversion (CARA) utility functions with risk-tolerance coefficient 1 defined over their final wealth.⁹ Prior to entering the asset market, they choose to belong to one of two trader types: “transparent traders” and “opaque traders.” By default, they are transparent traders, and if they want to switch to being opaque traders, they have to pay a cost $c > 0$. We let μ denote the fraction of opaque traders. We will interpret transparent traders and opaque traders as mutual funds and hedge funds, respectively.¹⁰ The cost c can be interpreted as the cost that hedge funds incur in developing complex trading strategies, which will be introduced shortly, net of the difference between hedge funds’ and mutual funds’ cost of reporting and complying with regulations.

Random variables $\tilde{\theta}_T$ and $\tilde{\theta}_O$ in (1) represent information that is observable to traders. Two interpretations can be given. First, $\tilde{\theta}_T$ is *public* information that is observable to all traders, while $\tilde{\theta}_O$ is *private* information that is observable only to opaque traders. This interpretation attributes the difference between $\tilde{\theta}_T$ and $\tilde{\theta}_O$ to the nature of information itself. That is, $\tilde{\theta}_T$ represents “transparent” information that can be easily obtained, while $\tilde{\theta}_O$ means “opaque” information that needs resources and skills to decrypt. Since opaque traders are relatively more sophisticated, it makes sense to assume that they are the ones who have

⁹ Risk tolerance is the inverse of risk aversion. The unit risk tolerance and the one-share endowment are a normalization used only for convenience.

¹⁰ Although our analysis is cast in terms of hedge funds and mutual funds, its implications can be more general. For example, opaque traders and transparent traders can be interpreted as institutional traders and individual traders, as individual traders may not understand how institutions trade.

the ability to develop the information $\tilde{\theta}_O$. Second, both $\tilde{\theta}_T$ and $\tilde{\theta}_O$ can be interpreted as *private* information that are observed by different traders: $\tilde{\theta}_T$ is observed only by transparent traders, while $\tilde{\theta}_O$ is observed only by opaque traders. This interpretation puts more emphasis on who owns the information. Both interpretations yield the same equilibrium in our model. This is because the price aggregates the information in $\tilde{\theta}_T$ and $\tilde{\theta}_O$ (see Equation (11)), and opaque traders who know the structure of opaque trading can infer $\tilde{\theta}_T$ from the price even if they do not observe it directly. We are agnostic with respect to these two interpretations, but for simplicity, we have adopted the first one in the exposition.

When the asset market opens, transparent traders observe the equilibrium stock price $\tilde{p}$ and the information $\tilde{\theta}_T$, and trade the bond and the stock. Opaque traders are different from transparent traders in two ways. First, they have a different information set—that is, they observe an extra signal $\tilde{\theta}_O$ (in addition to the equilibrium price $\tilde{p}$ and the information $\tilde{\theta}_T$). Second, they have an enlarged investment opportunity set (which is developed from the cost c that they spent)—that is, in addition to the bond and the stock, opaque traders can access additional investment opportunities that are not available to transparent traders. One can interpret these extra investment opportunities as, for example, venture capital, foreign currencies, and options, in which equity mutual funds typically do not invest.

We assume that all opaque traders have the same investment opportunities. Let $1 + \tilde{\eta}$ be the (gross) returns on these extra investment opportunities, where $\tilde{\eta} \sim N(0, \sigma_\eta^2)$ with $\sigma_\eta > 0$.¹¹ In addition, $\tilde{\varepsilon}$ and $\tilde{\eta}$ are correlated with a coefficient of $\rho \in (0, 1)$, so that opaque traders can use these investment opportunities to hedge stock investments. As we will explain shortly in Section 1.3, the structure of these investment opportunities, or more specifically, the correlation coefficient ρ , is the source of transparent traders' "opaque" perspective about the trading strategy of opaque traders in the stock market. This makes sense given that transparent traders have no experience of trading the extra investment opportunities.

1.2 Demand function of opaque traders

Since we are interested in how opaque trading affects prices and welfare, we first derive opaque traders' demand for the stock, which is viewed as being "opaque" by transparent traders. Let D_O and Z be opaque traders' investment in the stock and the private investment opportunities. Then, given a starting wealth of $(\tilde{p} - c)$ before trade, which is the value of their stock endowment minus the cost of switching type, their future wealth after trade is

$$\tilde{W}_O = (\tilde{p} - c) + (\tilde{v} - \tilde{p})D_O + Z\tilde{\eta}, \quad (2)$$

¹¹ For simplicity, we have assumed that the mean net return on extra investment opportunities is 0. In Appendix A, we show that our results are robust to an extension that relaxes this assumption.

and the CARA-normal setup implies that their expected utility is

$$E[-\exp(-\tilde{W}_O)|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O] = -\exp\left[-\left(E(\tilde{W}_O|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O) - \frac{1}{2}\text{Var}(\tilde{W}_O|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O)\right)\right], \quad (3)$$

where $E(\cdot|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O)$ and $\text{Var}(\cdot|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O)$ are the conditional mean and variance operators.

By Equation (1), we have

$$E(\tilde{W}_O|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O) = (\tilde{p} - c) + (\bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - \tilde{p})D_O, \quad (4)$$

$$\text{Var}(\tilde{W}_O|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O) = \sigma_\varepsilon^2 D_O^2 + \sigma_\eta^2 Z^2 + 2\rho\sigma_\varepsilon\sigma_\eta D_O Z. \quad (5)$$

Inserting these moment expressions into the objective function of opaque traders in (3) and solving their decision problem yields:

$$D_O^* = \frac{E(\tilde{v}|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O) - \tilde{p}}{(1 - \rho^2)\sigma_\varepsilon^2} \quad \text{and} \quad Z^* = -\frac{\rho[E(\tilde{v}|\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O) - \tilde{p}]}{(1 - \rho^2)\sigma_\varepsilon\sigma_\eta}. \quad (6)$$

We define

$$k \equiv \frac{1}{1 - \rho^2} > 1, \quad (7)$$

and so the demand function in Equation (6) can be written as

$$D_O(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k) = \frac{k(\bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - \tilde{p})}{\sigma_\varepsilon^2}. \quad (8)$$

Note that Equation (8) is identical to the demand function of a CARA investor who has a risk tolerance k . So, the presence of the additional investment opportunities causes opaque traders to behave *as if* they are more risk tolerant than transparent traders to the extent that the returns on these additional investment opportunities are correlated with the returns on traded assets. For this reason, we call parameter k the opaque traders' *effective* risk tolerance in the stock market, and explicitly incorporate it as an argument in the demand function (8).

We make two remarks regarding opaque traders' investment problem. First, for simplicity, we have assumed that all opaque traders have the same investment opportunities. This assumption aims to capture the idea that the main source of ambiguity for mutual funds is the nondiversifiable component of hedge fund trades that may come from their common investment opportunities. This assumption also implies that any hedge fund (opaque trader) in our model does not face ambiguity about the trading strategies of other hedge funds, since they all have the same parameter k , which is the source of ambiguity in our setup. Our motivation for this assumption is that hedge funds often know more than mutual funds know about what other hedge funds are doing—that is, since mutual funds and hedge funds have different institutional structures, incentives, and constraints, it is likely that hedge funds will tend to have a certain degree

of commonality in the kinds of strategic analysis and investigation that they do relative to mutual funds (and vice versa).

In effect, Equation (6) shows that the extra investment opportunities $\tilde{\eta}$ affect opaque traders' stock investments only through its correlation ρ with the stock. So, even if opaque traders face different investment opportunities, as long as the correlation of each such investment with the stock is the same, opaque traders' stock investment will be the same. In addition, in Appendix A, we show that our results are robust to an extension that allows opaque traders to access different types of investment opportunities (characterized by different correlations with the stock) so that opaque traders do not make identical stock investments.

Second, although we motivate the increased effective risk tolerance of opaque traders using an enlarged investment opportunity set, it could equally well come from differences in the institutional environment. For example, because the ability of investors to withdraw their investments from hedge funds is limited, hedge funds tend to have more liquidity than mutual funds. As a consequence, hedge funds will appear to be less risk averse in the asset market than mutual funds. So, henceforth, we work with this more general notation k in determining the investment opportunity structure.

1.3 Trading opacity

To capture the "opacity" of the trading strategy of opaque traders, we assume that transparent traders perceive ambiguity about the effective risk-tolerance parameter k of opaque traders; that is, transparent traders perceive that

$$k \in [\underline{k}, \bar{k}] \text{ with } 1 \leq \underline{k} \leq \bar{k}, \quad (9)$$

and they are unable to assign a probability on this set. At a more micro-level, this trading strategy ambiguity reflects transparent traders' uncertainty regarding the structure of the investment opportunity set (for example, the parameter ρ in the previous subsection) or liquidity conditions of opaque traders.

This ambiguity will cause stock prices, which incorporate opaque traders' information $\tilde{\theta}_O$ through their trading, to become an ambiguous signal about the stock payoff $\tilde{v}$. We use a boldface $\mathbf{k} \geq 1$ to denote the true value of k , and assume that $\underline{k}$ and $\bar{k}$ are generated as follows:

$$\underline{k} \equiv \mathbf{k} - \Delta k \text{ and } \bar{k} \equiv \mathbf{k} + \Delta k, \quad (10)$$

where $\Delta k \geq 0$ is an exogenous parameter that determines the amount of ambiguity faced by transparent traders.

Incorporating ambiguity is particularly appropriate when decision makers lack enough information or experience to assess the relevant distribution over payoffs (see Heath and Tversky 1991; Epstein and Schneider 2008). This aptly characterizes the situation with hedge funds because historically hedge funds were exempt from registration and various reporting obligations required of other investment entities such as mutual funds. The exemption was justified because qualified hedge fund investors were believed able to make informed

investment decisions without regulatory oversight. With limited transparency, however, it is difficult for investors not in hedge funds to know what is going on in the industry or to assess the relevant distributions of payoffs generated by the sophisticated trading strategies. For these traders, ambiguity prevails, and it would be expected to affect their behavior and that of the market. We show in Section 5 that our key results cannot be obtained in an asymmetric information model without ambiguity aversion.

The timeline of the model is given as follows. Initially, each trader is endowed with one share of the stock and decides whether to pay c to switch from being a transparent trader to an opaque trader in the asset market. Before the asset market opens, transparent traders observe information $\tilde{\theta}_T$, and opaque traders observe information $\{\tilde{\theta}_T, \tilde{\theta}_O\}$ and learn the structure of the additional investment opportunities (and hence, they realize their new effective risk-tolerance parameter k). Then the asset market opens and all traders trade the bond and the stock at prices 1 and $\tilde{p}$, respectively; opaque traders also invest in the additional investment opportunities. After trading ends, traders receive the payoffs on their portfolios and consume.

To summarize, the tuple

$$\mathcal{E} = (\bar{v}, \sigma_{\theta T}, \sigma_{\theta O}, \sigma_\varepsilon, \mathbf{k}, \Delta k, c)$$

defines an economy. We are interested in the implications for asset prices, trader distributions, and welfare of changes in the parameters Δk and c , which are in turn influenced by regulatory policies.

2. Equilibrium

We consider rational expectations equilibria (REE). We start by analyzing trading behavior and prices in the financial market, given some fraction μ of opaque traders. We then analyze traders' decisions about whether to become opaque or stay transparent, and determine the equilibrium fraction μ^* of opaque traders. We focus on economies with interior $\mu^* \in (0, 1)$; that is, ones in which both transparent and opaque traders are active in the asset market, as this is the empirically relevant case.

2.1 Financial market equilibrium

2.1.1 Characterization. For any given k , we assume that transparent traders rationally conjecture that the price function is:

$$\tilde{p} = \bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - f(k), \quad (11)$$

where the function $f(k)$ will be endogenously determined in equilibrium. Function $f(k)$ is also the transparent traders' *perceived* equity premium for a given belief k : $E_k(\bar{v} - \tilde{p})$, where $E_k(\cdot)$ denotes an expectation using the belief k .

Since transparent traders are ambiguous about k , they view the stock price $\tilde{p}$ as an ambiguous signal about $\tilde{\theta}_O$. Thus, the decision problem for transparent traders is

$$\max_{D_T} \min_{k \in [\underline{k}, \bar{k}]} E_k \left(-e^{-\tilde{W}_T} \middle| \tilde{p}, \tilde{\theta}_T \right), \quad (12)$$

subject to the standard budget constraint:

$$\tilde{W}_T = \tilde{p} + D_T(\tilde{v} - \tilde{p}), \quad (13)$$

where in Equation (12), $E_k(\cdot | \tilde{p}, \tilde{\theta}_T)$ denotes a conditional expectation given the belief k , and in Equation (13), the first term, $\tilde{p}$, is the value of transparent traders' stock endowment, and D_T is their demand for the stock. It follows immediately from our normal distribution structure, that the above decision problem is equivalent to:

$$\max_{D_T} \min_{k \in [\underline{k}, \bar{k}]} \left([E_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) - \tilde{p}] D_T - \frac{1}{2} \text{Var}_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) D_T^2 \right). \quad (14)$$

By Equation (11), $E_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T)$ and $\text{Var}_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T)$, the conditional moments of $\tilde{v}$ taken under a particular belief k , are given by:

$$E_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) = \tilde{p} + f(k) \quad \text{and} \quad \text{Var}_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) = \sigma_\varepsilon^2,$$

and as a result, for a fixed investment D_T , we have:

$$[E_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) - \tilde{p}] D_T - \frac{1}{2} \text{Var}_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) D_T^2 = -\frac{1}{2} \sigma_\varepsilon^2 D_T^2 + f(k) D_T.$$

Define the minimum and maximum values that $f(\cdot)$ takes on as

$$f_{\min} \equiv \min_{k \in [\underline{k}, \bar{k}]} f(k) \quad \text{and} \quad f_{\max} \equiv \max_{k \in [\underline{k}, \bar{k}]} f(k). \quad (15)$$

Then the objective function of a transparent trader can be written as

$$\begin{aligned} & \min_{k \in [\underline{k}, \bar{k}]} \left([E_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) - \tilde{p}] D_T - \frac{1}{2} \text{Var}_k(\tilde{v} | \tilde{p}, \tilde{\theta}_T) D_T^2 \right) \\ &= \begin{cases} -\frac{1}{2} \sigma_\varepsilon^2 D_T^2 + f_{\min} D_T, & \text{if } D_T \geq 0, \\ -\frac{1}{2} \sigma_\varepsilon^2 D_T^2 + f_{\max} D_T, & \text{if } D_T < 0. \end{cases} \end{aligned} \quad (16)$$

Thus a transparent trader's demand function is:

$$D_T(\tilde{p}, \tilde{\theta}_T) = \begin{cases} \frac{f_{\min}}{\sigma_\varepsilon^2}, & \text{if } f_{\min} > 0, \\ 0, & \text{if } f_{\min} \leq 0 \leq f_{\max}, \\ \frac{f_{\max}}{\sigma_\varepsilon^2}, & \text{if } f_{\max} < 0. \end{cases} \quad (17)$$

This demand function is very intuitive. Recall that function $f(k)$ represents the transparent traders' perceived equity premium for a given belief k . So, if the

minimum perceived equity premium across all possible beliefs is still positive, then transparent traders will be confident that the stock is undervalued relative to its fundamentals, and hence they will buy the stock. Similarly, if the maximum perceived equity premium is negative, they are confident that the stock is overvalued relative to its fundamentals, so that they will go short the stock. In contrast, when the perceived equity premium is positive for some beliefs but negative for others, then transparent traders are not sure about whether the stock is under- or overvalued, and to be conservative in their risky trading positions (as ambiguity-averse traders), they will not hold any stock.

Using the two demand functions, we can determine the functional form of $f(k)$ from the market-clearing condition

$$\mu D_O(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k) + (1 - \mu) D_T(\tilde{p}, \tilde{\theta}_T) = 1. \quad (18)$$

The transparent trader's demand $D_T(\tilde{p}, \tilde{\theta}_T)$ can take the three possible values in Equation (17), so we will discuss market clearing case by case.

First, suppose $D_T(\tilde{p}, \tilde{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$. Substituting this expression into the market-clearing condition (Equation (18)), we can solve for the equilibrium price $\tilde{p}$:

$$\tilde{p} = \bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - \frac{-(1 - \mu) f_{\min} + \sigma_\varepsilon^2}{\mu k}.$$

Comparing this price function with the conjectured price function in Equation (11), we see that if the transparent traders' conjecture is rational, then:

$$f(k) = \frac{-(1 - \mu) f_{\min} + \sigma_\varepsilon^2}{\mu k}. \quad (19)$$

In the current case in which $D_T(\tilde{p}, \tilde{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$ in Equation (17), we must have $f_{\min} > 0$, and hence $f(k) > 0$ for all possible k . So $f(k)$ achieves its minimum at $\bar{k}$ in Equation (19); that is,

$$f_{\min} = \frac{-(1 - \mu) f_{\min} + \sigma_\varepsilon^2}{\mu \bar{k}} \Rightarrow f_{\min} = \frac{\sigma_\varepsilon^2}{1 - \mu + \mu \bar{k}}. \quad (20)$$

Substituting this value back into Equation (19) and solving for the function f , we have

$$f(k) = \frac{\bar{k} \sigma_\varepsilon^2}{k(1 - \mu + \mu \bar{k})}. \quad (21)$$

Second, suppose that $D_T(\tilde{p}, \tilde{\theta}_T) = 0$. Then, the market-clearing condition implies $f(k) = \frac{\sigma_\varepsilon^2}{\mu k}$. However, this means that $f_{\min} > 0$, which is inconsistent with the case $D_T(\tilde{p}, \tilde{\theta}_T) = 0$, as from Equation (17) we see that $D_T(\tilde{p}, \tilde{\theta}_T) = 0$ implies that $f_{\min} \leq 0$. Similarly, we can eliminate the case in which the transparent traders' demand is $\frac{f_{\max}}{\sigma_\varepsilon^2}$.

The argument above is summarized in the following proposition, which describes the rational expectations equilibrium.

Proposition 1. Suppose $0 < \mu < 1$. There exists an REE in which the price function is

$$\tilde{p} = \bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - f(k),$$

where

$$f(k) = \frac{\bar{k}\sigma_\varepsilon^2}{k(1 - \mu + \mu\bar{k})}.$$

2.1.2 Implications for the equity premium. The equity premium of the stock is defined as:

$$EP \equiv \mathbf{E}(\tilde{v} - \tilde{p}) = \frac{(\mathbf{k} + \Delta k)\sigma_\varepsilon^2}{\mathbf{k}[(1 - \mu) + \mu(\mathbf{k} + \Delta k)]}, \quad (22)$$

where the boldfaced $\mathbf{E}(\cdot)$ indicates that the expectation is taken under the true value $\mathbf{k}$ of the parameter k , and where the second equality follows from Equation (21) and the definition of $\bar{k}$ in Equation (10).

Clearly, for any fixed $\mu \in (0, 1)$, we have:

$$\frac{\partial EP}{\partial \Delta k} > 0;$$

that is, a decrease in the amount of ambiguity will decrease the equity premium. This occurs because in equilibrium, transparent traders must hold some of the stock (because of the aggregate positive stock supply), and if they are less uncertain about what opaque traders are doing, for any given price, they are more optimistic about the payoff of the stock and hence require a lower equity premium to be compensated for holding it. Formally, we have the following corollary.

Corollary 1. When $\mu \in (0, 1)$ is fixed, reducing the amount of ambiguity will lower the equity premium.

This result has important policy implications. Requiring hedge funds to disclose their positions more frequently can be interpreted in our model as reducing Δk , as more frequent disclosure helps the market to understand what is going on in the black box of the hedge fund industry. Note that disclosure does not have to be interpreted as eliminating ambiguity; instead, disclosure may reduce ambiguity—that is, reduce Δk —without necessarily driving it to zero.¹²

¹² Specifically, imagine the following dynamic setting where the effective risk tolerance k_t of opaque traders evolves according to the following process: $k_t = ak_{t-1} + b_t$. Here, $a \in (0, 1)$ is a known constant, and b_t is an ambiguous innovation term to transparent traders: they perceive $b_t \in [\underline{b}, \bar{b}]$ and cannot assign a probability on this set. If there is no disclosure about opaque traders' previous positions, then transparent traders have no idea of the realizations of k_{t-j} , and as a result, at date t , transparent traders would view k_t to be possible on the whole set of $\left[\frac{\underline{b}}{1-a}, \frac{\bar{b}}{1-a}\right]$. If, in contrast, opaque traders disclose their positions with one period lag, then transparent traders can use Equations (8) and (11) to back out k_{t-1} , so that at date t , their ambiguity set about k_t shrinks to a smaller set $[k_{t-1} + \underline{b}, k_{t-1} + \bar{b}] \subset \left[\frac{\underline{b}}{1-a}, \frac{\bar{b}}{1-a}\right]$.

Thus, implementing this practice would tend to lower the equity premium in the short run when the number of hedge funds active in the market has not been affected (i.e., when μ is fixed).

2.1.3 The ex post performance of opaque traders. Empirical studies find that actively managed stock mutual funds as a group do not perform better than the stock market, while hedge funds beat mutual funds after adjusting for beta risk (i.e., the risk arising from exposure to common market movements).¹³ In this section we show that this difference in returns occurs in equilibrium in our model. This is not surprising as hedge funds managers in our economy have an information advantage and are actively developing additional investment opportunities that produce alpha returns.

When we interpret the tradable risky asset as the whole stock market, the return on the market portfolio in excess of the interest rate (which is normalized to 1 in our model) is

$$\tilde{R}_M = \frac{\tilde{v}}{\tilde{p}} - 1. \quad (23)$$

Substituting the relevant branch of the transparent traders demand function, that is, $D_T(\tilde{p}, \tilde{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$, into their budget constraint, and dividing by their initial wealth $\tilde{p}$ shows that the excess return on the equilibrium portfolio of transparent traders is:

$$\tilde{R}_T = \frac{\tilde{W}_T}{\tilde{p}} - 1 = \frac{f_{\min}}{\sigma_\varepsilon^2} \tilde{R}_M. \quad (24)$$

An econometrician with access to historical data would compute the beta of transparent traders' portfolio as:

$$\beta_T \equiv \frac{\mathbf{Cov}(\tilde{R}_T, \tilde{R}_M | \tilde{p})}{\mathbf{Var}(\tilde{R}_M | \tilde{p})} = \frac{f_{\min}}{\sigma_\varepsilon^2}, \quad (25)$$

where the boldfaced $\mathbf{Cov}(\cdot | \tilde{p})$ and $\mathbf{Var}(\cdot | \tilde{p})$ indicate that the conditional moments are taken under the true value $\mathbf{k}$ of the parameter k . We assume that the beta is computed conditional on price $\tilde{p}$ because in practice, price information is indeed available to econometricians.

Thus, by Equations (23)–(25), the market-adjusted returns on transparent traders' equilibrium portfolio is

$$\alpha_T \equiv \mathbf{E}(\tilde{R}_T | \tilde{p}) - \beta_T \mathbf{E}(\tilde{R}_M | \tilde{p}) = 0.$$

This result is intuitive: since transparent traders hold only the risk-free bond with a zero rate of return and the market portfolio, their portfolio returns do not beat the market.

¹³ See Stulz (2007) for a survey on the performance of hedge funds and mutual funds.

Inserting the optimal investments in the tradable assets and the additional investment opportunities (Equation (6)) into the opaque traders' budget constraint (Equation (2)) and dividing by their total invested capital ($\tilde{p} - c$) shows that the excess return on the equilibrium portfolio of opaque traders is:

$$\tilde{R}_O = \frac{\tilde{W}_O}{\tilde{p} - c} - 1 = \frac{1}{\tilde{p} - c} \left[\frac{(\tilde{\varepsilon} + f(k))f(k)}{(1 - \rho^2)\sigma_\varepsilon^2} - \frac{\rho f(k)}{(1 - \rho^2)\sigma_\varepsilon\sigma_\eta} \tilde{\eta} \right]. \quad (26)$$

In Appendix B, we use Equations (23) and (26) to show that an outside econometrician would compute the beta and alpha of opaque traders' portfolio as:

$$\beta_O \equiv \frac{\text{Cov}(\tilde{R}_O, \tilde{R}_M | \tilde{p})}{\text{Var}(\tilde{R}_M | \tilde{p})} = \frac{\tilde{p}}{\tilde{p} - c} \frac{EP}{\sigma_\varepsilon^2}, \quad (27)$$

$$\alpha_O \equiv \mathbf{E}(\tilde{R}_O | \tilde{p}) - \beta_O \mathbf{E}(\tilde{R}_M | \tilde{p}) = \frac{\mathbf{k} - 1}{\tilde{p} - c} \frac{EP^2}{\sigma_\varepsilon^2}, \quad (28)$$

where EP is the equity premium of the stock given by Equation (22). Since $\mathbf{k} > 1$, opaque traders appear to generate positive alphas in our economy, as long as the net wealth ($\tilde{p} - c$) of opaque traders is positive.

2.2 Trader distribution equilibrium

We now analyze traders' choices of whether to remain transparent or pay the cost c and become opaque. Whether a transparent trader would like to switch depends on the comparison between the ex ante expected utility of becoming opaque and that of staying transparent.

Inserting the optimal investments of an opaque trader (Equation (6)) into Equation (3), and using the moment expressions of Equations (4)–(5) and the definition of k in Equation (7), we see that the indirect utility of an opaque trader in a market with price $\tilde{p}$ is

$$V_{O1}(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k) = -\exp \left[- \left(\tilde{p} - c + \frac{k}{2\sigma_\varepsilon^2} (\bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - \tilde{p})^2 \right) \right], \quad (29)$$

and the ex ante expected utility of becoming opaque is

$$V_{O0} = \min_{k \in [\underline{k}, \bar{k}]} E_k [V_{O1}(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k)].$$

Using the equilibrium price function in Proposition 1, we have

$$V_{O0} = -\exp \left(- \left[\bar{v} - c + \min_k \left[-f(k) + k \frac{[f(k)]^2}{2\sigma_\varepsilon^2} \right] - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right] \right). \quad (30)$$

Similarly, substituting the relevant branch of demand function of transparent traders, that is, $D_T(\tilde{p}, \tilde{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$, into their budget constraint and objective

function shows that the indirect utility of being a transparent trader in a market with price $\tilde{p}$ is

$$V_{T1}(\tilde{p}, \tilde{\theta}_T) = \min_{k \in [\underline{k}, \bar{k}]} E_k \left[-e^{-(\tilde{p} + D_T(\tilde{p}, \tilde{\theta}_T)(\bar{v} - \tilde{p}))} \middle| \tilde{p}, \tilde{\theta}_T \right] = -\exp \left[-\left(\tilde{p} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right) \right], \quad (31)$$

where the second equality follows from Equation (16). Given the recursive multiple-priors utility representation, the ex ante expected utility of staying transparent is

$$V_{T0} = \min_{k \in [\underline{k}, \bar{k}]} E_k [V_{T1}(\tilde{p}, \tilde{\theta}_T)] \quad (32)$$

$$= -\exp \left[-\left(\bar{v} - f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right) \right] \quad (33)$$

$$= -\exp \left[-\left(\bar{v} + \min_k [-f(k)] + \min_k \frac{[f(k)]^2}{2\sigma_\varepsilon^2} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right) \right], \quad (34)$$

where (33) follows from (31) and the equilibrium price function in Proposition 1 and where (34) simply rewrites (33) more explicitly to facilitate interpretation.

Note that the perceived equity premium $f(k)$ plays two different roles in affecting the ex ante expected utilities V_{O0} and V_{T0} . First, a high $f(k)$ represents a high discount on the expected wealth level associated with traders' unit stock endowment, thereby reducing ex ante utilities. This role is reflected by the linear term $-f(k)$ in (30) and (34). Second, a high $f(k)$ also represents a high perceived benefit resulting from future trading in the asset market. When the equity premium is high, the stock price, on average, will deviate from the stock's liquidation value by a large amount, generating attractive trading opportunities for traders. This second role of the equity premium is captured by the quadratic terms $k \frac{[f(k)]^2}{2\sigma_\varepsilon^2}$ and $\frac{[f(k)]^2}{2\sigma_\varepsilon^2}$ in (30) and (34), respectively.

Opaque and transparent traders evaluate these two roles of the equity premium $f(k)$ differently in Equations (30) and (34). Specifically, both types of traders face ambiguity about k ex ante, and so when they evaluate the two roles of $f(k)$, they will employ minimum operators to reflect their ambiguity aversion. However, in (30), opaque traders apply one minimum operator jointly to the linear and quadratic terms of $f(k)$, $\min_k \left[-f(k) + k \frac{[f(k)]^2}{2\sigma_\varepsilon^2} \right]$, while transparent traders employ two separate minimum operators in (34), $\min_k [-f(k)] + \min_k \frac{[f(k)]^2}{2\sigma_\varepsilon^2}$. This difference reflects the fact that transparent traders face ambiguity both at the type-decision stage and at the trading stage (and hence the two minimum operators in (34)), while opaque traders face ambiguity only at the type-decision stage and resolve their ambiguity at the trading stage (as it is their aggregate trading behavior that creates the

ambiguity). This difference also offers an advantage to traders who become opaque that is related to our ambiguity-aversion approach. Intuitively, the strength of this advantage is positively related to the amount Δk of ambiguity. We show in Section 5 that this advantage is critical for our results about the effect of hedge fund regulation.

By Equations (30) and (33), the benefit of switching from being a transparent trader to an opaque trader as a function of the fraction μ of opaque traders is:

$$B(\mu) \equiv \min_k \left[-f(k) + k \frac{[f(k)]^2}{2\sigma_\varepsilon^2} \right] - \left(-f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right). \quad (35)$$

In order for an interior fraction $\mu^* \in (0, 1)$ of opaque traders to be an equilibrium, it must be that every transparent trader is indifferent between becoming opaque versus remaining transparent—that is, $B(\mu^*) = c$. In Appendix B we show that such an equilibrium exists when c takes values in a range whose size is positively associated with the degree Δk of ambiguity of transparent traders about the trading behavior of opaque traders. This is intuitive, since as discussed above, switching to being an opaque trader can bring a benefit through removing the ambiguity at the trading stage, and the size of this benefit is positively related to Δk .

In addition, we show that the benefit function $B(\cdot)$ is decreasing in μ , because increasing μ increases the ex ante expected utility of a transparent trader but decreases that of an opaque trader. Specifically, increasing μ reduces the equity premium on average since opaque traders trade more aggressively than transparent traders. As noted above, the reduced equity premium will benefit traders ex ante through increasing their expected wealth, but harm them through decreasing their trading benefits. Since transparent traders trade less aggressively than opaque traders in the financial market, the trading benefits channel affects their ex ante utility to a smaller extent. As a result, the net effect of increasing μ is to benefit transparent traders and harm opaque traders. Hence, the benefit to a transparent trader of becoming opaque decreases with μ , and any interior fraction of opaque traders ($0 < \mu^* < 1$) must be unique.¹⁴

Formally, we have the following proposition regarding equilibrium fraction μ^* of opaque traders (with proof provided in Appendix B).

Proposition 2. Let $\underline{c} \equiv \left(\frac{1}{\bar{k}} - \frac{1}{\bar{k}^2} \right) \frac{\sigma_\varepsilon^2}{2}$, $\bar{c} \equiv \left(\min \left\{ \frac{\bar{k}^2}{\bar{k}} - 1, \frac{2\bar{k}}{\bar{k}} + \bar{k} - 3 \right\} \right) \frac{\sigma_\varepsilon^2}{2}$. When $\underline{c} < c < \bar{c}$, there exists a unique interior trader distribution equilibrium $0 < \mu^* < 1$. In addition, the size of the range $(\underline{c}, \bar{c})$ increases with the degree of ambiguity; that is, $\frac{\partial(\bar{c} - \underline{c})}{\partial \Delta k} > 0$.

¹⁴ This is in contrast to Mele and Sangiori (2011), who show that introducing exogenous ambiguity about cash flows in a standard Grossman and Stiglitz (1980) model can make the function $B(\cdot)$ become increasing in μ . What drives the difference is the feature of changing traders' risk tolerance k in our model, which is absent in Mele and Sangiori (2011). The role of ambiguity in our model is to strengthen the effect of this feature.

3. Regulation, Prices, and Welfare

In this section, we discuss the implications of changes in the cost, c , of becoming opaque and changes in the extent of ambiguity, Δk , for the equilibrium trader distribution μ^* , the equity premium EP^* , and welfare. We focus on changes in c and Δk because these parameters relate to current policy debates. The recent SEC registration requirement under the Dodd-Frank Act for hedge funds can be understood to imply an increase in the cost c of operating a hedge fund.¹⁵ The suggestion that hedge funds disclose their positions to the public (with some delay) implies a decrease in Δk . Note that for the economies we consider, those with interior $\mu^* \in (0, 1)$, welfare can be measured by the expected utility of transparent traders, since all traders are ex ante identical and in equilibrium they all have the same welfare.

3.1 Implication of registration: c

In this subsection we examine the impact of increasing c while keeping Δk fixed. We begin by analyzing the effect on the equilibrium fraction μ^* of opaque traders, which is determined by the condition $B(\mu^*) = c$. Note that, the benefit, $B(\cdot)$, of switching types is affected only by ambiguity parameters $\underline{k}$ and $\bar{k}$, and not by the cost c . As a result, an increase in c will reduce the incentive of transparent traders to become opaque, leading to fewer opaque traders—that is, a lower μ^* . If c increases μ^* must decrease to restore the equality $B(\mu^*) = c$.

In Panels (a1)—(a3) of Figure 1 we use a numerical example to illustrate this result. We take one period as one year and suppose the cost increases from 0.05 to 0.1, while other relevant parameter values are fixed as follows: $\sigma_\varepsilon = 0.2$, $\mathbf{k} = 2$, and $\Delta k = 1$. The choice of $\sigma_\varepsilon = 0.2$ aims to match the 20% historical volatility of aggregate stock returns, which facilitates our interpretation of the equity premium result.¹⁶ The choices for other parameters are made in a relatively ad hoc manner, but our results are robust to alternative choices. Panel (a1) plots the benefit function $B(\cdot)$ and two horizontal cost functions. For any cost, the equilibrium fraction μ^* of opaque traders is determined by the intersection of function $B(\cdot)$ and the horizontal cost function. We see that increasing c from 0.05 to 0.1 decreases μ^* from 0.39 to 0.07.

The equilibrium equity premium is given by Equation (22) evaluated at the equilibrium μ^* :

$$EP^* = \frac{(\mathbf{k} + \Delta k)\sigma_\varepsilon^2}{\mathbf{k}[1 + \mu^*(\mathbf{k} - 1) + \mu^*\Delta k]}. \quad (36)$$

Because opaque traders have a greater effective risk-tolerance coefficient (the term $\mu^*(\mathbf{k} - 1)$ in the denominator of the above equation) and do not face ambiguity (the term $\mu^*\Delta k$ in the denominator of the above equation), they

¹⁵ See Sjostrom (2011) for a discussion on the history of hedge funds registration with SEC.

¹⁶ Leland (1992) has made the same choice in a one-period model for studying insider trading.

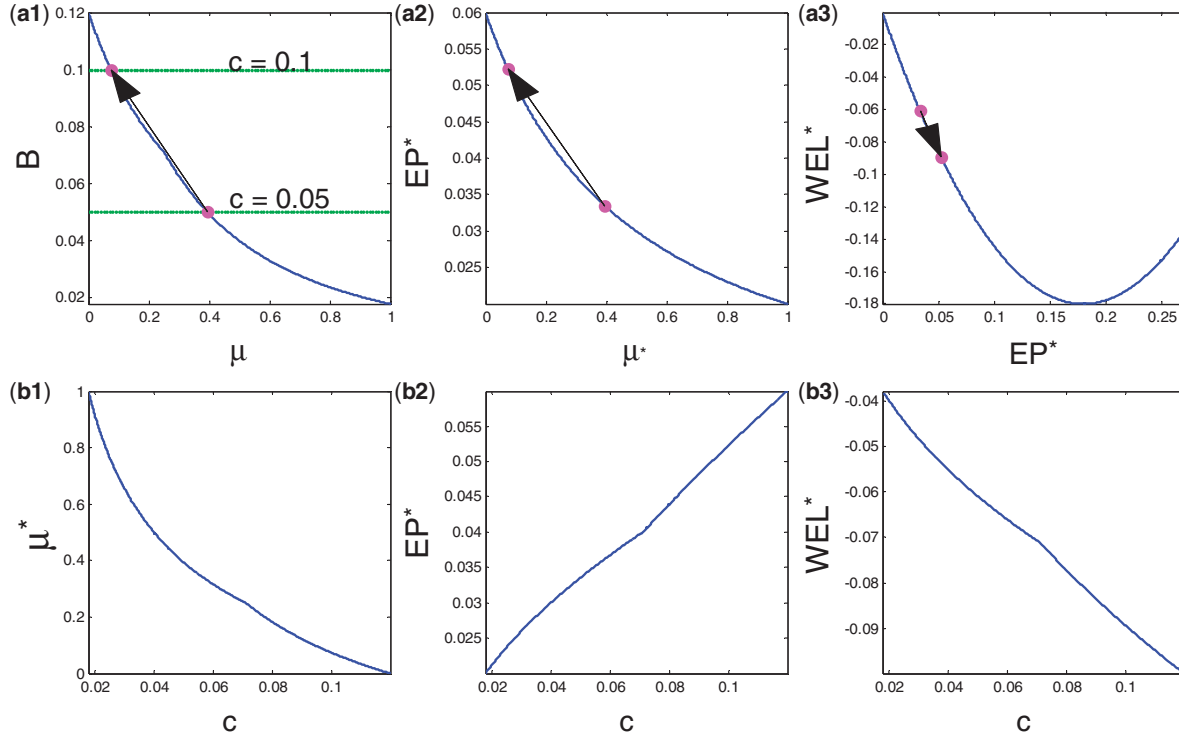


Figure 1

Implications for equilibrium market outcomes of increasing c

This figure shows the impact of increasing the cost c of switching trader types on the equilibrium fraction μ^* of opaque traders, the equity premium EP^* , and welfare WEL^* . In the upper panels, c is assumed to increase from 0.05 to 0.1 when the other parameters take values of $\sigma_E = 0.2$, $k = 2$, and $\Delta k = 1$. Panel (a1) plots benefit function B , and the equilibrium fraction μ^* of opaque traders is determined by the intersection of function B and the horizontal cost functions. Panel (a2) plots equilibrium equity premium EP^* as a function of μ^* . Panel (a3) plots equilibrium welfare WEL^* as a function of the equilibrium equity premium EP^* . The three lower panels plot μ^* , EP^* , and WEL^* as functions of c , respectively, when the other parameters take values of $\sigma_E = 0.2$, $k = 2$, and $\Delta k = 1$.

trade more aggressively than transparent traders. Therefore, a reduction in μ^* due to an increase in c will lower asset prices and increase the equity premium. We illustrate this effect by continuing the numerical example. Panel (a2) of Figure 1 plots Equation (36), and it shows that the drop in μ^* increases the equilibrium equity premium from 3.35% to 5.23%.

We measure welfare by the certainty equivalent of transparent traders' ex ante equilibrium utility (adjusted by subtracting a constant $\bar{v} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2}$), which, by Equation (33), is

$$WEL^* = -f_{\max}^* + \frac{f_{\min}^{*2}}{2\sigma_{\varepsilon}^2}, \quad (37)$$

where $f_{\max}^*$ and $f_{\min}^*$ are transparent traders' perceived maximum and minimum equilibrium equity premium (given by Equation (15) evaluated at μ^*). These two variables capture the two different roles of the equity premium in affecting traders' ex ante utility: an increase in $f_{\max}^*$ reduces welfare, because it reflects transparent traders' perceived discount of their expected wealth, while an increase in $f_{\min}^*$ increases welfare, because it captures the perceived benefit resulting from future trading in the asset market.¹⁷

Given the definitions of $f_{\max}^*$, $f_{\min}^*$ and the true equity premium EP^* , we know that $f_{\max}^* = \frac{k}{k} EP^*$ and $f_{\min}^* = \frac{k}{k} EP^*$. Substituting these expressions into Equation (37) provides the following relationship between equilibrium welfare WEL^* and the equity premium EP^* :

$$WEL^* = -\frac{k}{k} EP^* + \frac{(k/\bar{k})^2}{2\sigma_{\varepsilon}^2} EP^{*2}. \quad (38)$$

The first term $-\frac{k}{k} EP^*$ shows that a high EP^* tends to reduce welfare through its impact on wealth, while the second term $\frac{(k/\bar{k})^2}{2\sigma_{\varepsilon}^2} EP^{*2}$ shows that a high EP^* tends to increase welfare through trading. The net effect of an increase in EP^* on welfare depends on the relative strength of these two terms. In our economy, the first term dominates, so that when the equity premium becomes larger, welfare will decrease.¹⁸ Thus, an increase in the cost of becoming an opaque trader, c , increases the equity premium, EP^* , and reduces welfare, WEL^* . Continuing the example in Figure 1, we see in its Panel (a3), which plots Equation (38), that the increase in WEL^* decreases welfare from -0.06 to -0.09 .

The results in this subsection are summarized in the following proposition.

¹⁷ Note that transparent traders perceive the equity premium differently at the trading stage and at the trader type-decision stage. This difference does not mean that they are time inconsistent, since the belief construction in our economy satisfies Epstein and Schneider (2003)'s "rectangularity condition" and the preference has a recursive structure. Instead, it simply reflects the fact that transparent traders face ambiguity at both stages.

¹⁸ To see this, notice that when Δk is fixed, $\frac{\partial WEL^*}{\partial EP^*} < 0$ if and only if $EP^* < \frac{k^2 \sigma_{\varepsilon}^2}{k\bar{k}}$, which is true for any $0 < \mu^* < 1$ by Equation (36).

Proposition 3. Suppose $0 < \mu^* < 1$. An increase in c will decrease the fraction μ^* of opaque traders, increase the equity premium EP^* , and decrease welfare WEL^* .

The lower panels of Figure 1 use the same numerical example as its upper panels to illustrate Proposition 3. Indeed, as we gradually increase c , μ^* decreases, EP^* increases, and WEL^* decreases.

3.2 Implication of disclosure: Δk

In this subsection we examine the impact of decreasing the amount of ambiguity, Δk , while keeping c fixed. A decrease in Δk will reduce the benefit of switching from being a transparent trader to being an opaque trader, as the reduced Δk weakens the advantage of opaque traders not facing ambiguity when trading the stock. As a consequence, the equilibrium fraction μ^* of opaque traders decreases as Δk decreases.

Panels (a1)–(a3) of Figure 2 use an example similar to the economy in Figure 1 to illustrate this result. We decrease Δk from 1 to 0.5, while other relevant parameters are fixed as follows: $\sigma_\varepsilon = 0.2$, $k = 2$, and $c = 0.05$. Panel (a1) plots benefit functions $B(\cdot)$ when $\Delta k = 1$ (solid curve) and when $\Delta k = 0.5$ (dashed curve). The equilibrium fraction μ^* of opaque traders is determined by the intersection of the benefit functions and the horizontal cost function. We find that decreasing Δk from 1 to 0.5 shifts the benefit function downward, leading to a decrease in μ^* from 0.39 to 0.06.

The impact of a change in Δk on the equity premium EP^* is complex because, according to Equation (36), Δk affects the equity premium in two opposing ways. First, as captured by the numerator of Equation (36) and formalized by Corollary 1, when μ^* is fixed, a decrease in Δk will directly decrease the equity premium, because transparent traders face less uncertainty in interpreting the information in the price and hence require a lower equity premium to hold the stock. Second, there is an indirect effect, as captured by the denominator of Equation (36), which works through the endogenous μ^* : a decrease in Δk causes μ^* to decrease, which, in turn, tends to increase the equity premium, since opaque traders trade more aggressively than transparent traders. The total effect of decreasing Δk on EP^* depends on the relative strength of these two effects.

In Appendix B we show that a sufficient condition for the second effect to dominate is:

$$\mu^* > \frac{(\bar{k}/2) - 1}{\bar{k} - 1}. \quad (39)$$

That is, if μ^* is sufficiently large, then decreasing Δk will increase the equity premium. To see why this is true, consider the extreme case in which μ^* is close to 1. In this case, since almost all traders are opaque traders and face no ambiguity, varying Δk has almost no direct effect on the price. However, as μ^*

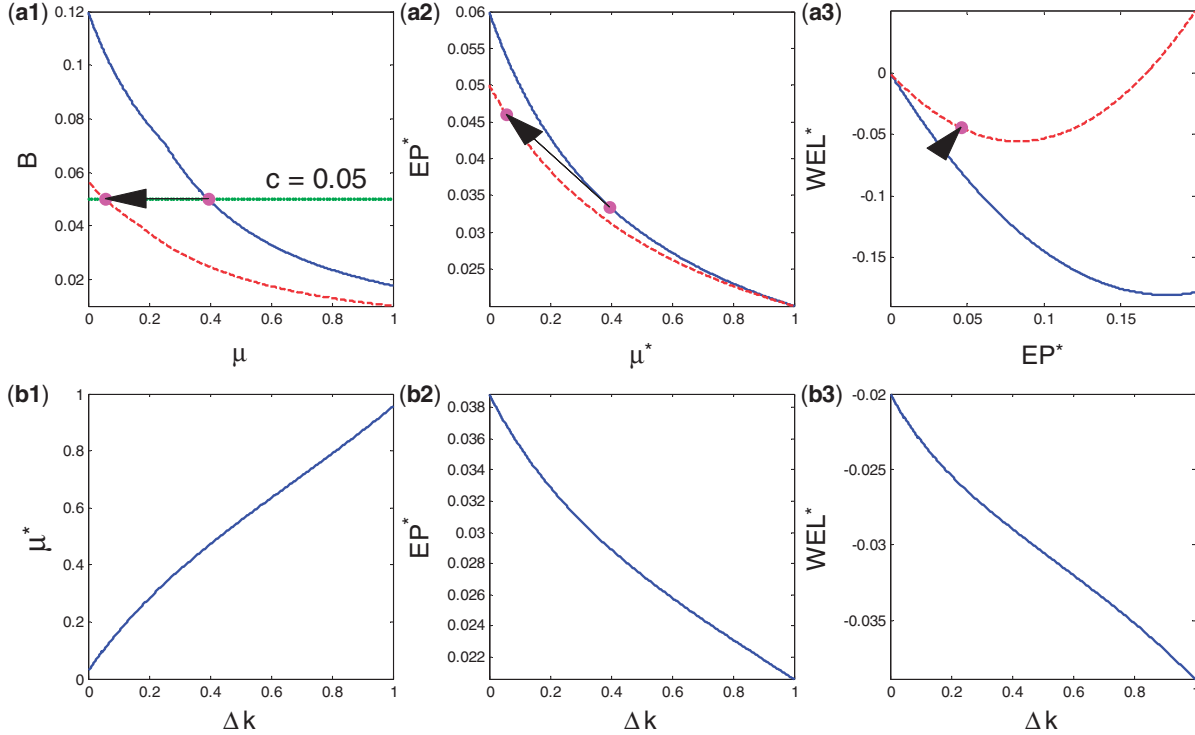


Figure 2
Implications for equilibrium market outcomes of decreasing Δk

This figure shows the impact of decreasing the degree of ambiguity Δk on the equilibrium fraction μ^* of opaque traders, the equity premium EP^* , and welfare WEL^* . In the upper panels, Δk is assumed to decrease from 1 to 0.5 when the other parameters take values of $\sigma_e = 0.2$, $k = 2$, and $c = 0.05$. Panel (a1) plots benefit functions B , and the equilibrium fraction μ^* of opaque traders is determined by the intersection of functions B and the horizontal cost function. Panel (a2) plots equilibrium equity premium EP^* as functions of μ^* . Panel (a3) plots equilibrium welfare WEL^* as functions of the equilibrium equity premium EP^* . The lower panels plot μ^* , EP^* , and WEL^* as functions of Δk when other parameters take values of $\sigma_e = 0.2$, $k = 2$, and $c = 0.019$.

begins to decline due to a decrease in Δk , the indirect effect is still effective, which lowers the price and increases the equity premium.

The range in (39) is large. In fact, when $\bar{k} \leq 2$, the lower bound $\frac{(\bar{k}/2)-1}{\bar{k}-1}$ is negative, and thus for all interior values of $\mu^* \in (0, 1)$, a decrease in Δk will raise the equity premium, EP^* . Also, note that the range includes all values of $\mu^* > \frac{1}{2}$, since the lower bound $\frac{(\bar{k}/2)-1}{\bar{k}-1}$ is smaller than $\frac{1}{2}$. So a more stringent disclosure requirement on hedge fund positions may initially lower the equity premium (as predicted by Corollary 1 for a fixed μ), but the equity premium will eventually increase as the fraction of hedge funds adjusts to its new equilibrium value.

We illustrate these effects by continuing our example in Figure 2. Panel (a2) plots Equation (36) when $\Delta k = 1$ (solid curve) and when $\Delta k = 0.5$ (dashed curve). Consistent with Corollary 1, a decrease in Δk shifts the curve downward, showing that when μ^* is fixed at its initial value of 0.39, decreasing Δk will reduce the equity premium EP^* . However, decreasing Δk also reduces μ^* from 0.39 to 0.06, and this decreased μ^* causes the equity premium, EP^* , to increase from 3.35% to 4.61%.

Unlike the previous subsection where we saw that changing c always affects the equity premium and welfare in opposite ways, it is possible for a decrease in Δk to improve welfare while increasing the equity premium. This can occur because when ambiguity-averse traders evaluate their welfare, they tend to overestimate the effect of equity premium in discounting expected wealth and to underestimate the effect of equity premium in increasing the trading benefit. This happens because in Equation (38), the coefficient $\frac{k}{\bar{k}}$ is greater than one and the coefficient $\frac{(k/\bar{k})^2}{2\sigma_\varepsilon^2}$ is less than $\frac{1}{2\sigma_\varepsilon^2}$. As a result, when disclosure reduces ambiguity, so that Δk decreases, their estimates of k are closer to the true value, which improves traders' welfare ex ante. When this positive effect is strong enough to dominate the potential negative effect of the increased equity premium, traders actually are made better off by more disclosure. Appendix B provides a sufficient condition for a decrease in Δk to improve equilibrium welfare: $\bar{k} < \frac{\bar{k}}{2} \left(\sqrt{1+2\bar{k}} - 1 \right)$.

Panel (a3) of Figure 2 illustrates this point. Here, we plot welfare WEL^* as a function of the equity premium EP^* (i.e., Equation (38)) when $\Delta k = 1$ (solid curve) and when $\Delta k = 0.5$ (dashed curve). We find that decreasing Δk from 1 to 0.5 moves the whole curve upward, meaning that if the equity premium is fixed at its initial value, 3.35%, then disclosure will improve welfare from -0.061 to -0.037 . However, decreasing Δk also raises the equity premium from 3.35% to 4.61%, and as a result, welfare does not improve as much as it would for a fixed equity premium; that is, WEL^* increases from -0.061 only to -0.044 . Still, in total, in this example disclosure is welfare-improving.

The following proposition summarizes the results in this subsection (with proof given in Appendix B):

Proposition 4. Suppose $0 < \mu^* < 1$. A decrease in Δk will reduce the fraction μ^* of opaque traders. If $\mu^* > \frac{(\bar{k}/2)-1}{\bar{k}-1}$, then a decrease in Δk will increase the equity premium EP^* , and if, in addition, $\bar{k} < \frac{\bar{k}}{2} \left(\sqrt{1+2\bar{k}} - 1 \right)$, then a decrease in Δk will increase welfare WEL^* .

Although Proposition 4 only provides sufficient conditions for the increase in welfare, this result holds in various numerical examples. Panels (b1)–(b3) of Figure 2 provide a typical example with parameter values $\sigma_\varepsilon = 0.2$, $\mathbf{k} = 2$, and $c = 0.019$, where the choice of c is made to ensure an interior μ^* for all possible values of Δk in the range of $[0, 1]$. These panels show that a decrease in Δk consistently decreases μ^* and increases EP^* and WEL^* .

4. The Role of Information Asymmetry

Two features of our model are important in driving our analysis of the effects of regulatory policy: (i) information asymmetry—that is, opaque traders have private information that is also useful to transparent traders, and (ii) ambiguity about opaque trading—that is, transparent traders face ambiguity about how opaque traders trade on their information and transparent traders are ambiguity averse. In the next two sections, we show that both features are crucial for the endogenous ambiguity-averse trading decisions of transparent traders, which in turn, are crucial for our results about the effects of regulatory policy.

Specifically, in this section, we investigate the role of information asymmetry by analyzing an economy, $\text{Economy}^{\text{sym}}$, in which opaque traders do not have private signals, but transparent traders are ambiguous about the trading strategies of opaque traders. Thus, comparing this $\text{Economy}^{\text{sym}}$ with the economy in our main model isolates the role of information asymmetry in delivering our results. We show that, in $\text{Economy}^{\text{sym}}$, transparent traders do not learn from prices, and they trade stocks as traditional expected utility maximizers even though there is ambiguity about hedge fund strategies and transparent traders are ambiguity averse. This occurs because transparent traders do not directly care about hedge fund strategies. They only care about what hedge funds know about the value of the stock—and in this economy hedge funds do not have any private information about the value of the stock. More generally, in a market equilibrium no trader directly cares about any other trader's investment demands. The only way that any other trader's demand affects another trader is through its effect on the equilibrium price, which every trader knows.

In the next section, we investigate the role of ambiguity about the opaque trading parameter k . To do so, we assume that opaque traders have a private signal and then extend our main model to allow traders to have preferences

represented by “smooth ambiguity aversion” (Klibanoff, Marinacci, and Mukerji 2005). A special case of this extension will shut down ambiguity by assuming that transparent traders have a prior over the possible values of k . We show that, in this special case (and in cases close to it in terms of preferences), many of our results related to Δk are reversed.

We first analyze Economy^{sym} and show that information asymmetry is critical for our results. We set $\sigma_{\theta O} = 0$ in Economy^{sym}, so that opaque traders’ private signal $\tilde{\theta}_O$ is reduced to its unconditional mean 0, and all information available in the economy is the public information $\tilde{\theta}_T$. Note that the price function in Proposition 1 does not depend on the parameter $\sigma_{\theta O}$, suggesting that we cannot simply take the limit of $\sigma_{\theta O}$ in our main model to get the results in Economy^{sym}. This occurs because transparent traders in Economy^{sym} do not need to infer information from prices to make decisions and so the form of their demand function is different. Thus we re-derive the equilibrium for Economy^{sym}.

In Economy^{sym} the opaque and transparent traders’ demand functions for the stock are, respectively,

$$D_O(\tilde{p}^{sym}, \tilde{\theta}_T) = \frac{k(\bar{v} + \tilde{\theta}_T - \tilde{p}^{sym})}{\sigma_\varepsilon^2} \text{ and } D_T(\tilde{p}^{sym}, \tilde{\theta}_T) = \frac{\bar{v} + \tilde{\theta}_T - \tilde{p}^{sym}}{\sigma_\varepsilon^2}. \quad (40)$$

In particular, the transparent traders’ demand function has the standard form obtained from the utility maximization problem of a traditional Walrasian CARA investor. This is in stark contrast to the demand function (17) in our main model. The reason for this difference is that in Economy^{sym}, transparent traders know information $\{\tilde{p}^{sym}, \tilde{\theta}_T\}$, and the only randomness in their future wealth $\tilde{W}_T = \tilde{p}^{sym} + D_T(\tilde{v} - \tilde{p}^{sym})$ is the element ε in the stock payoff. Since no one in the economy has information about ε , the price $\tilde{p}^{sym}$ does not contain any information about it, and transparent traders do not use the price $\tilde{p}^{sym}$ to update their beliefs. So, unlike our main model, here, the ambiguity about parameter k does not affect transparent traders’ portfolio decisions.

Substituting the above demand functions into the market-clearing condition, $\mu D_O(\tilde{p}^{sym}, \tilde{\theta}_T) + (1 - \mu) D_T(\tilde{p}^{sym}, \tilde{\theta}_T) = 1$, we see that the functional form of the price $\tilde{p}^{sym}$ is

$$\tilde{p}^{sym} = \bar{v} + \tilde{\theta}_T - f^{sym}(k), \quad (41)$$

where

$$f^{sym}(k) = \frac{\sigma_\varepsilon^2}{1 - \mu + \mu k} \quad (42)$$

is the perceived equity premium for a given belief k in Economy^{sym}.

The equity premium of the stock in this economy is

$$EP^{sym} \equiv \mathbf{E}(\tilde{v} - \tilde{p}^{sym}) = \frac{\sigma_\varepsilon^2}{1 - \mu + \mu \mathbf{k}}. \quad (43)$$

Comparing Equations (22) and (43), we immediately notice two features that differentiate Economy^{sym} from our main model. First, in Economy^{sym},

the opacity measure Δk does not directly affect the equity premium EP^{sym} . Consequently, information asymmetry is important for delivering our earlier finding in Corollary 1 that disclosure policies designed to reduce opacity do affect the equity premium. Second, for any given μ , the equity premium in Economy^{sym} is smaller than it is in our main model; that is, $EP^{sym} < EP$. This is because information asymmetry, combined with the fact that transparent traders are ambiguous about k , creates its own risk for the transparent traders and lowers their demand in the economy studied in our main model.

We now derive the benefit of switching from being a transparent trader to an opaque trader. In Appendix B, we show that the ex ante expected utility of staying transparent is

$$V_{T0}^{sym} = -\exp \left[- \left(\bar{v} - f_{\max}^{sym} + \frac{1}{2} \frac{(f_{\max}^{sym})^2}{\sigma_\varepsilon^2} - \frac{\sigma_{\theta T}^2}{2} \right) \right], \quad (44)$$

where

$$f_{\max}^{sym} \equiv \max_{k \in [\underline{k}, \bar{k}]} f^{sym}(k) = \frac{\sigma_\varepsilon^2}{1 - \mu + \mu \bar{k}}. \quad (45)$$

In Equation (44), the linear and quadratic terms of $f_{\max}^{sym}$, respectively, capture the two roles of the perceived equity premium on traders' ex ante utility—discounting the expected wealth level associated with the unit stock endowment and determining the trading benefit of traders as stock holders in the financial market. Unlike Equation (33), now transparent traders use the same perceived equity premium, $f_{\max}^{sym}$, in both terms of Equation (44). This is because at the trading stage, observing the price $\tilde{p}^{sym}$ enables transparent traders to infer the value of k . This difference also suggests that one important role of the private signal received by opaque traders is to shield the parameter k from being revealed to the transparent traders.

The ex ante expected utility of becoming opaque is more complicated. But, we can show that when $\mu > \frac{1}{2}$, it takes the following simple form:

$$V_{O0}^{sym} = \exp \left[- \left(\bar{v} - c - f_{\max}^{sym} + \frac{k}{2\sigma_\varepsilon^2} (f_{\max}^{sym})^2 - \frac{\sigma_{\theta T}^2}{2} \right) \right]. \quad (46)$$

Again, the term $-f_{\max}^{sym}$ is the discount applied to the expected wealth level, and the term $\frac{k}{2\sigma_\varepsilon^2} (f_{\max}^{sym})^2$ captures the trading benefit. Therefore, in the region of $[1/2, 1]$, the benefit $B^{sym}(\mu)$ of switching trader types is:

$$B^{sym}(\mu) = \frac{k-1}{2\sigma_\varepsilon^2} (f_{\max}^{sym})^2, \quad (47)$$

which is the difference in the trading benefit of the two types of traders. Function $B^{sym}(\mu)$ is still downward sloping, so that the equilibrium fraction μ^* of opaque traders is unique.

Unlike in our main model, where decreasing Δk always lowers the benefit function B , decreasing Δk has two opposing effects on B^{sym} in Economy^{sym}. First, by Equation (45), it will decrease the maximum perceived equity premium $f_{\max}^{sym}$, and thus decrease the trading benefit for each unit of stock purchased. Second, it will increase the conservative estimation of the extra trading benefit associated with the more aggressive trading of opaque traders, and this effect is captured by the term $\frac{k-1}{2\sigma_\varepsilon^2}$ in Equation (47). As a result of these two opposing effects, decreasing Δk does not necessarily lower the benefit B^{sym} .¹⁹

These results are very different from those in Proposition 4, which shows that decreasing Δk always decreases μ^* and increases the equity premium (for $\mu^* > \frac{1}{2}$) in our main model with information asymmetry.²⁰ Intuitively, in our main model, because opaque traders have an extra information advantage whose strength is positively related to Δk , lowering Δk will lower the benefit generated by this information advantage, and as a result, decreasing Δk will always shift the curve $B(\mu)$ downward. We believe that this information asymmetry is more in line with reality, as hedge funds often claim that disclosure harms their interests through revealing their trading secrets and reducing their information advantage (cf., Aragon, Hertz, and Shi, 2012).

5. The Role of Ambiguity Aversion: Smooth Ambiguity Aversion

In this section, we extend our analysis to allow traders to have preferences represented by “smooth ambiguity aversion” as characterized by Klibanoff, Marinacci, and Mukerji (2005), while keeping all of the other features the same as in our baseline model. This extension serves two purposes. First, as mentioned in the previous section, it nests the traditional subjective expected utility preference (Savage (1954); hereafter SEU), which can work as a benchmark to pinpoint the role of ambiguity aversion in delivering our results. We show that the most interesting results about Δk can be obtained only with sufficient ambiguity aversion; for example, Corollary 1 and Proposition 4 can be reversed in an economy without ambiguity aversion. Second, this extension also nests our baseline model with the “kinked” max-min ambiguity aversion, and so we can rely on it to demonstrate the robustness of our results by showing that they continue to hold as long as traders’ ambiguity aversion is sufficiently large, although still “smooth.”

¹⁹ Specifically, we can show that this is true if and only if $\mu^* > \frac{1}{k-1}$. That is, if $\mu^* > \frac{1}{k-1}$, decreasing Δk will decrease B^{sym} , and hence decrease μ^* and increase $E P^{sym}$ (by Equation (43)); if $\frac{1}{2} < \mu^* < \frac{1}{k-1}$, decreasing Δk will increase B^{sym} , and hence increase μ^* and decrease $E P^{sym}$.

²⁰ Note that the condition of $\mu^* > \frac{(\bar{k}/2)-1}{k-1}$ in Proposition 4 includes the range $[1/2, 1]$.

5.1 Preference specification and REE characterization

For any given belief k , transparent traders still rationally conjecture a price function given by Equation (11). Under smooth ambiguity aversion, a transparent trader chooses stock demand D_T to maximize

$$E_Q[\phi(E_k[-e^{-(\bar{p}+D_T(\bar{v}-\bar{p}))}|\tilde{p},\tilde{\theta}_T])|\tilde{p},\tilde{\theta}_T], \quad (48)$$

subject to the budget constraint (13).

Here, Q is a probability measure on k and it represents the “model uncertainty” (where the word “model” refers to a statistical model parameterized by belief k). $E_Q[\cdot]$ is the expectation operator with respect to the measure Q . For simplicity, we adopt a uniform measure Q ; that is, under measure Q , k is uniformly distributed over its support $[\underline{k}, \bar{k}]$. The size of k ’s support is still $\bar{k} - \underline{k} = 2\Delta k$, which is typically interpreted as the objective amount of ambiguity.

The function $\phi(\cdot)$ represents the trader’s subjective attitude toward ambiguity. It transforms the conditional expected utility given the belief k — $E_k[-e^{-(\bar{p}+D_T(\bar{v}-\bar{p}))}|\tilde{p},\tilde{\theta}_T]$ —into a real number, and it is strictly increasing and continuous (and hence the name “smooth ambiguity aversion”). In particular, if $\phi(\cdot)$ is strictly concave, the trader is ambiguity averse. We follow the literature (e.g., Caskey 2009; Ju and Miao 2012) and specify a “constant relative ambiguity aversion function”:

$$\phi(u) = -(-u)^a, \text{ with } a \geq 1, \quad (49)$$

where a indexes the trader’s ambiguity aversion coefficient. In particular, a value $a = 1$ corresponds to the traditional SEU preference (Savage 1954), and a value $a = \infty$ corresponds to our baseline model with max-min ambiguity (Gilboa and Schmeidler 1989).

Given specification (48), direct computations show that a transparent trader’s optimal demand for the stock D_T^* is given as follows:

$$D_T^* = E_Q\left[\frac{f(k)}{\sigma_\varepsilon^2}\right] + Cov_Q\left(\frac{e^{-aD_T^*f(k)}}{E_Q[e^{-aD_T^*f(k)}]}, \frac{f(k)}{\sigma_\varepsilon^2}\right), \quad (50)$$

where $Cov_Q(\cdot, \cdot)$ is the covariance operator with respect to measure Q . In the above expression, the first term is an average (over k) stock demand $\frac{f(k)}{\sigma_\varepsilon^2}$ of a CARA investor who perfectly knows k (recall that $f(k)$ is the per-unit equity premium and σ_ε^2 is the residual randomness in the stock cash flow). The second term is a covariance term, which is negative and reflects a downward adjustment in the demand for the stock due to ambiguity aversion.

Since opaque traders do not face ambiguity at the trading stage, their demand function and indirect utility function are still given by Equations (8) and (29), respectively. Following the same procedure as in our baseline model, we substitute these demand functions into the market-clearing condition (18) to

solve for the actual price function, and then compare it with the conjectured price function, yielding:

$$f(k) = \frac{-(1-\mu)\sigma_\varepsilon^2 D_T^* + \sigma_\varepsilon^2}{\mu k}. \quad (51)$$

Equations (50) and (51) jointly characterize the financial market equilibrium. Specifically, inserting (51) into (50) delivers the following equation in terms of the single unknown constant D_T^* :

$$D_T^* = E_Q \left[\frac{-(1-\mu)D_T^* + 1}{\mu k} \right] + Cov_Q \left(\frac{e^{-aD_T^*} \frac{-(1-\mu)\sigma_\varepsilon^2 D_T^* + \sigma_\varepsilon^2}{\mu k}}{E_Q \left[e^{-aD_T^*} \frac{-(1-\mu)\sigma_\varepsilon^2 D_T^* + \sigma_\varepsilon^2}{\mu k} \right]}, \frac{-(1-\mu)D_T^* + 1}{\mu k} \right). \quad (52)$$

We can use the intermediate value theorem to show that there exists a solution D_T^* to the above equation and that any solution D_T^* is in the range $(0, 1)$. We can then easily numerically compute D_T^* , and then insert the solved D_T^* into (51) to obtain $f(k)$ and hence the price function (11). We summarize these results in the following proposition.

Proposition 5. Suppose $0 < \mu < 1$. There exists an REE in the economy with smooth ambiguity aversion, and the price function is

$$\tilde{p} = \tilde{v} + \tilde{\theta}_T + \tilde{\theta}_O - f(k),$$

where $f(k) = \frac{-(1-\mu)\sigma_\varepsilon^2 D_T^* + \sigma_\varepsilon^2}{\mu k}$ with D_T^* being given by (52). In addition, $D_T^* \in (0, 1)$ and $f(k) > 0$.

5.2 Pricing implications of disclosure at the trading stage

We next revisit Corollary 1, which examines the implication of reducing Δk for the equity premium— $EP \equiv \mathbf{E}(\tilde{v} - \tilde{p}) = f(\mathbf{k})$ —when the fraction μ of opaque traders is exogenous. Our analysis suggests that Corollary 1 continues to hold as long as the ambiguity-attitude parameter a is sufficiently large, so that transparent traders' preference is close to the specification in our baseline model. Interestingly, we also find that Corollary 1 is reversed when a is small—that is, when transparent traders can be approximately characterized by the traditional SEU preference. This suggests that ambiguity aversion is crucial in delivering Corollary 1.

The complexity of (52) precludes a full analytical characterization. We therefore conduct a numerical analysis in Panels (a1)–(a3) of Figure 3. Here, we continue to follow our previous figures and set $\sigma_\varepsilon = 0.2$ and $\mathbf{k} = 2$. The exogenous μ is fixed at 0.5, and our results are robust to this choice. We report results for three values of ambiguity-attitude parameter a —1, 50, and

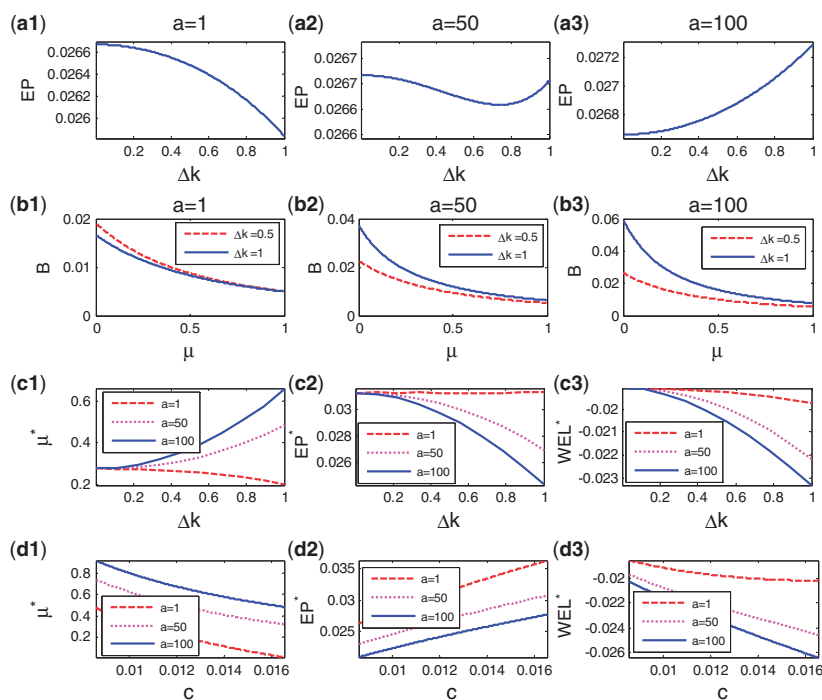


Figure 3

Implications of registration and disclosure in economies with smooth ambiguity

This figure assumes that traders have smooth ambiguity preferences. Parameter a controls traders' ambiguity attitude: a higher a means that traders are more ambiguity averse. When $a = 1$, traders are Savage investors, and when $a = \infty$, traders have max-min preferences. In all panels, $\sigma_\varepsilon = 0.2$ and $k = 2$. Panels (a1)–(a3) examine the implications of changing Δk for the equity premium when the fraction of opaque traders is fixed at $\mu = 0.5$. Panels (b1)–(b3) plot benefit functions B when Δk can take values of either 0.5 or 1. Panels (c1)–(c3) plot μ^* , EP^* , and WEL^* as functions of Δk when c is set at 0.012. Panels (d1)–(d3) plot those variables as functions of c when Δk is fixed at 1.

100—which are chosen to reflect all possible patterns. In particular, the case of $a = 1$ corresponds to the SEU preference without ambiguity aversion, and the case of $a = 100$ is similar to the max-min ambiguity aversion as specified in our baseline model. We see that Corollary 1 holds in Panel (a3) with a large value of a : the equity premium EP increases with the size of Δk , which is a proxy for the amount of ambiguity and may be reduced by more frequent disclosure. In contrast, in Panel (a1) with $a = 1$ (SEU), the equity premium decreases with Δk , and in Panel (a2) with $a = 50$, the equity premium first decreases and then increases with Δk .

These contrasting results suggest that there are two competing forces driving the relation between the equity premium, EP , and the amount of objective ambiguity, Δk . One force dominates in the case of the max-min ambiguity aversion, as in our baseline model, and the other dominates in the case of SEU

preferences. Note that by Equation (51), the equity premium is given by

$$EP = \frac{-(1-\mu)\sigma_\varepsilon^2 D_T^* + \sigma_\varepsilon^2}{\mu k};$$

that is, the equity premium is negatively affected by the equilibrium demand D_T^* of transparent traders. By Equation (50), D_T^* is in turn determined by two terms: $E_Q\left[\frac{f(k)}{\sigma_\varepsilon^2}\right]$ and $Cov_Q\left(\frac{e^{-aD_T^* f(k)}}{E_Q[e^{-aD_T^* f(k)}]}, \frac{f(k)}{\sigma_\varepsilon^2}\right)$. The amount of objective ambiguity Δk affects these two terms differently, leading to the two competing forces. Increasing Δk tends to increase the term $E_Q\left[\frac{f(k)}{\sigma_\varepsilon^2}\right]$, which reflects the fact that parameter k nonlinearly affects $f(k)$ in equilibrium. Specifically, by (51), $f(k)$ is a convex function of k , and hence increasing Δk tends to increase $E_Q\left[\frac{f(k)}{\sigma_\varepsilon^2}\right]$ through a Jensen's inequality effect. The term $Cov_Q\left(\frac{e^{-aD_T^* f(k)}}{E_Q[e^{-aD_T^* f(k)}]}, \frac{f(k)}{\sigma_\varepsilon^2}\right)$ is negative, and it reflects the downward adjustment on D_T^* due to ambiguity aversion. So, the size of this covariance term increases with the objective amount of ambiguity Δk and/or the subjective ambiguity aversion coefficient a . When the ambiguity-attitude parameter a is sufficiently high, this second effect dominates, so that increasing Δk reduces transparent traders' demand and increases EP , as highlighted by our baseline model with $a = \infty$.

5.3 Characterization of trader distribution equilibrium

Smooth ambiguity aversion admits a recursive structure that we use to determine ex ante utility and thus the value of becoming an opaque trader. We use the opaque traders' indirect utility expression in (29) and compute the ex ante expected utility of becoming opaque as follows:

$$\begin{aligned} V_{O0} &= \phi^{-1}\left(E_Q\left[\phi\left(E_k\left[V_{O1}(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k)\right]\right)\right]\right) \\ &= -\exp\left(-\left[\tilde{v} - c - \frac{1}{a} \log\left(E_Q\left[e^{-a\left(-f(k) + \frac{k}{2\sigma_\varepsilon^2}[f(k)]^2\right)}\right]\right) - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2}\right]\right), \end{aligned} \quad (53)$$

where $\phi^{-1}(\cdot)$ is the inverse function of $\phi(\cdot)$. The indirect utility of a transparent trader at the trading stage is given by

$$V_{T1}(\tilde{p}, \tilde{\theta}_T) = \phi^{-1}\left(E_Q\left[\phi\left(E_k\left[-e^{-(\tilde{p} + D_T^*(\tilde{v} - \tilde{p}))}\right]\tilde{p}, \tilde{\theta}_T\right)\right]\tilde{p}, \tilde{\theta}_T\right]$$

and we can compute the ex ante utility of staying transparent as:

$$\begin{aligned} V_{T0} &= \phi^{-1} \left(E_Q \left[\phi \left(E_k \left[V_{T1}(\tilde{p}, \tilde{\theta}_T) \right] \right) \right] \right) \\ &= -\exp \left[- \left(\bar{v} - \frac{1}{a} \left[\log \left(E_Q [e^{af(k)}] E_Q [e^{\frac{a}{2} \sigma_\varepsilon^2 D_T^{*2} - a D_T^* f(k)}] \right) \right] - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right) \right]. \end{aligned} \quad (54)$$

Thus, the benefit of switching to being an opaque trader is:

$$\begin{aligned} B(\mu) &= \frac{1}{a} \left[\log \left(E_Q [e^{af(k)}] E_Q [e^{\frac{a}{2} \sigma_\varepsilon^2 D_T^{*2} - a D_T^* f(k)}] \right) \right. \\ &\quad \left. - \log \left(E_Q \left[e^{-a \left(-f(k) + \frac{k}{2\sigma_\varepsilon^2} [f(k)]^2 \right)} \right] \right) \right]. \end{aligned} \quad (55)$$

The equilibrium fraction μ^* of opaque traders is still determined by $B(\mu) = c$, and the equilibrium welfare WEL^* is still given by the certainty equivalent of a transparent trader's ex ante equilibrium utility (adjusted by a constant $\bar{v} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2}$). We can show that as $a \rightarrow \infty$, V_{O0} , V_{T0} , and $B(\mu)$, respectively, converge to their counterparts in our baseline model.

5.4 Implications of disclosure and registration

We next examine the implications of changing Δk and c for μ^* , EP^* , and WEL^* when μ is endogenous (as in Propositions 3 and 4). We start with Δk . In Panels (b1)–(b3) of Figure 3, we continue our previous example and plot the benefit functions $B(\cdot)$ for two possible values of $\Delta k = 0.5$ and 1. We see that, in Panels (b2) and (b3) where the ambiguity-attitude parameter a is sufficiently large, the result is identical to that in our baseline model (Panel (a1) of Figure 2)—that is, reducing Δk from 1 to 0.5 moves downward the whole benefit function. However, we also see that in Panel (b1) with $a = 1$ (SEU preference), the result is opposite to that in our baseline model—that is, reducing Δk shifts upward the benefit function. This again highlights the fundamental difference of our ambiguity-aversion approach.

Since the benefit of switching types is given by the differences between traders' ex ante utilities V_{O0} and V_{T0} , we next examine what these differences are and how they are affected by Δk and a . First, as in our baseline model, opaque traders face ambiguity only once at the type-decision stage, while transparent traders face ambiguity twice at both the type-decision and trading stages. So, opaque traders use the single expectation term $E_Q [e^{-a(-f(k) + \frac{k}{2\sigma_\varepsilon^2} [f(k)]^2)}]$ in (53) to evaluate the two roles of $f(k)$ in discounting wealth (the linear term $-f(k)$ in the exponent of the integrand) and in generating trading benefits (the squared term $[f(k)]^2$ in the exponent). By contrast, in (54), transparent traders have to employ two expectation terms— $E_Q [e^{af(k)}]$ and $E_Q [e^{\frac{a}{2} \sigma_\varepsilon^2 D_T^{*2} - a D_T^* f(k)}]$ —to evaluate the separate roles of the

perceived equity premium $f(k)$ in discounting wealth and in generating trading benefits in the financial market.

Note that the two roles of $f(k)$ work against each other: a high $f(k)$ indicates a high discount of wealth and hurts traders, but at the same time, a high $f(k)$ also means a large deviation between the stock price and its fundamentals, which creates good trading opportunities and improves traders' welfare. This means that evaluating the two roles simultaneously tends to reduce the uncertainty due to k , and ambiguity-averse traders value this uncertainty reduction very much (given the concave function ϕ). Thus, in this way, opaque traders' facing ambiguity only once offers them an edge over transparent traders. This effect resembles the diversification effect in the traditional risk-aversion framework. For this reason, we call it the "ambiguity-diversification effect," and note that it is strong when traders are very ambiguity averse (a high a) and/or there is a large amount of ambiguity (a high Δk). This effect extends the intuition of our baseline model (with $a = \infty$) that increasing Δk will increase the incentive to become opaque.

The second difference between V_{O0} and V_{T0} arises because an opaque trader may become more risk tolerant at the trading stage (either because of a relaxed liquidity constraint or because of an augmented investment opportunity set as micro-founded in Section 1). This difference is captured by the presence of the term $\frac{k}{2\sigma_\varepsilon^2}$ in $E_Q[e^{-a(-f(k) + \frac{k}{2\sigma_\varepsilon^2}[f(k)]^2)}]$ in (53). The increased risk tolerance offers opaque traders another edge by making them better able to take advantage of trading opportunities. We label this effect the "risk-tolerance-shift effect." It is strong when Δk and/or a are small. This is because a smaller Δk makes traders ex ante more certain about the advantage generated by changing k at the future trading stage, and a smaller a means that traders can better use this advantage by reacting more strongly to it.²¹ This effect is also present in our baseline model with $a = \infty$, but it is dominated by the ambiguity-diversification effect.

In sum, Δk affects B through two effects: the ambiguity-diversification effect and the risk-tolerance-shift effect. When a is large, the first effect is strong and dominates. Since it increases with Δk , a reduction in Δk weakens the ambiguity-diversification effect and moves downward the B curves, as in Panels (b2) and (b3) of Figure 3. By contrast, when a is small, the risk-tolerance-shift effect dominates, and since a small Δk strengthens this effect, we see that a reduction in Δk moves upward the B curves in Panel (b1) of Figure 3.

Next we continue our analysis in Panels (c1)–(c3) of Figure 3 to examine how Δk affects the equilibrium fraction μ^* of opaque traders, the equity premium EP^* , and the welfare WEL^* . We still set $\sigma_\varepsilon = 0.2$ and $k = 2$. We then set $c = 0.012$

²¹ This risk-tolerance-shift effect resembles a result in Admati and Pfleiderer (1987), who argue that more risk-tolerant traders value information to a larger extent in a traditional risk-aversion framework.

to ensure that the equilibrium fraction μ^* of opaque traders takes an interior value for all possible values of Δk in the range of $[0, 1]$. For the cases of $a = 50$ and 100, we see that the results in Proposition 4 continue to hold: a decrease in Δk reduces μ^* , and increases EP^* and WEL^* .

However, for the case of $a = 1$ (SEU preference), the implications are different. First, decreasing Δk actually increases μ^* in Panel (c1). This directly follows from Panel (b1), which shows that decreasing Δk increases SEU traders' incentive to become opaque through the risk-tolerance-shift effect. Second, in Panel (c2), EP^* almost remains constant for the case of $a = 1$. This is because reducing Δk has two offsetting effects on the aggregate stock demand: on the one hand, it increases the fraction of opaque traders, who demand more of the stock; on the other hand, according to Panel (a1), the reduction in Δk also causes incumbent transparent traders to demand less of the stock, and the interaction of these two competing effects causes the overall price effect of reducing Δk to be mild. Finally, in Panel (c3), reducing Δk continues to improve traders' welfare for the case of $a = 1$, because even SEU investors benefit from uncertainty reduction.

We finally use Panels (d1)–(d3) of Figure 3 to examine the implications of changing the cost c of switching trader types. We find that Proposition 3 continues to hold in the economy with smooth ambiguity aversion. Specifically, for all three values of a , increasing c decreases μ^* , increases EP^* , and decreases WEL^* . The intuition is also the same as our baseline model.

6. Conclusion

We demonstrate the effect of “opaque trading” of hedge funds on asset prices and aggregate welfare. We employ an ambiguity-aversion approach to capture what we believe is an important aspect of hedge fund opacity: it is not just that hedge funds may have private information that is not available to other traders (information asymmetry), but rather that other traders are not even sure how hedge funds trade on their private information (ambiguity about trading strategies). In particular, this uncertainty about trading strategies introduces a variety of complications not captured by simple models of asymmetric information. We use our model to investigate the implications of regulations affecting the cost of operating a hedge fund or disclosure requirements influencing the amount of ambiguity about hedge fund strategies.

We find that increasing the differential cost of operating a hedge fund decreases the equilibrium fraction of hedge funds, increases the equity premium, and decreases welfare. Thus, regulatory policies designed simply to limit the number of hedge funds by increasing their costs seem ill advised. Increased disclosure requirements, however, have different effects because they can reduce the ambiguity about hedge fund strategies. The reduction in ambiguity reduces the attractiveness of becoming a hedge fund, and so results in fewer hedge funds in equilibrium. But the effect on the equity premium is

now more complicated. Reducing ambiguity makes non-hedge-funds traders more willing to trade, and this tends to reduce the equity premium, but as it reduces the number of hedge funds it reduces the fraction of aggressive traders in the market, and this tends to increase the equity premium. Similarly, the effect on welfare is ambiguous. It thus remains an empirical question whether reducing ambiguity has the effects that regulators desire.

Appendix A: Extensions to multidimensional ambiguity

In this Appendix, we extend our model to allow transparent traders to be ambiguous in multiple dimensions and show that our results continue to hold. We generate multidimensional ambiguity in two ways. In the first section below, transparent traders are ambiguous about multiple parameters in the structure of the opaque traders' additional investment opportunity. In the second section, opaque traders can access two types of additional investment opportunities, and transparent traders are ambiguous about both the structures.

Multidimensional ambiguity about the private investment opportunities. Opaque traders still have the same investment opportunity $\tilde{\eta}$. But we now allow for a non-zero mean of $\tilde{\eta}$; that is, $\tilde{\eta} \sim N(\bar{\eta}, \sigma_{\tilde{\eta}}^2)$ with $\bar{\eta} \in \mathbb{R}$ and $\sigma_{\tilde{\eta}} > 0$. We continue to assume that $\tilde{\varepsilon}$ and $\tilde{\eta}$ are correlated with a coefficient of $\rho \in (0, 1)$. transparent traders can potentially be ambiguous about all three parameters $(\rho, \bar{\eta}, \sigma_{\tilde{\eta}})$ that are related to opaque traders' investment opportunity. Other features of the economy are kept the same as those in our baseline model.

In this economy, we can show that opaque traders' demand for the stock and their indirect utility function are:

$$D_O^* = \frac{1}{1-\rho^2} \left(\frac{\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p}}{\sigma_{\tilde{\varepsilon}}^2} \right) - \frac{\rho}{(1-\rho^2)\sigma_{\tilde{\varepsilon}}} \left(\frac{\bar{\eta}}{\sigma_{\tilde{\eta}}} \right), \quad (\text{A1})$$

$$V_{O1}^* = -\exp \left\{ - \left[\begin{aligned} &\bar{p} - c + \frac{1}{2(1-\rho^2)} \left(\frac{\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p}}{\sigma_{\tilde{\varepsilon}}} \right)^2 \\ &- \frac{\rho}{1-\rho^2} \left(\frac{\bar{\eta}}{\sigma_{\tilde{\eta}}} \right) \left(\frac{\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p}}{\sigma_{\tilde{\varepsilon}}} \right) + \frac{1}{2(1-\rho^2)} \left(\frac{\bar{\eta}}{\sigma_{\tilde{\eta}}} \right)^2 \end{aligned} \right] \right\}. \quad (\text{A2})$$

The above expressions show that the extra investment opportunities affect D_O^* and V_{O1}^* only through parameters ρ and $s \equiv \frac{\bar{\eta}}{\sigma_{\tilde{\eta}}}$, where the ratio s is the Sharpe ratio of the extra investment opportunities. After redefining the effective risk-tolerance parameter $k \equiv \frac{1}{1-\rho^2}$ as in our baseline model and collecting parameters k and s into the vector

$$x \equiv (k, s), \quad (\text{A3})$$

we can rewrite D_O^* and V_{O1}^* as follows:

$$D_O(\bar{p}, \bar{\theta}_T, \bar{\theta}_O; x) = \frac{k(\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p})}{\sigma_{\tilde{\varepsilon}}^2} - \frac{s\sqrt{k(k-1)}}{\sigma_{\tilde{\varepsilon}}}, \quad (\text{A4})$$

$$V_{O1}(\bar{p}, \bar{\theta}_T, \bar{\theta}_O; x) = -\exp \left\{ - \left[\begin{aligned} &\bar{p} - c + \frac{k}{2\sigma_{\tilde{\varepsilon}}^2} (\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p})^2 \\ &- \frac{s\sqrt{k(k-1)}}{\sigma_{\tilde{\varepsilon}}} (\bar{v} + \bar{\theta}_T + \bar{\theta}_O - \bar{p}) + \frac{k s^2}{2} \end{aligned} \right] \right\}. \quad (\text{A5})$$

Equation (A4) implies that ambiguity about the three parameters $(\rho, \bar{\eta}, \sigma_{\tilde{\eta}})$ can be effectively captured by ambiguity about the two parameters (s, k) , as these parameters capture transparent

traders' ambiguity about the trading strategies $D_O(\bar{p}, \bar{\theta}_T, \bar{\theta}_O; x)$ of opaque traders. Similar to our baseline model, we assume that transparent traders perceive

$$k \in [\underline{k}, \bar{k}] \text{ and } s \in [\underline{s}, \bar{s}], \text{ with } 1 \leq \underline{k} \leq \bar{k} \text{ and } \underline{s} \leq 0 \leq \bar{s}, \quad (\text{A6})$$

and cannot assign a probability on this set. We impose the assumption of $\underline{s} \leq 0 \leq \bar{s}$ to nest our baseline model, which corresponds to the case of $\underline{s} = \bar{s} = 0$. This assumption is also realistic, because in reality, hedge funds could buy either projects with strong fundamentals (which corresponds to $\bar{s} \geq 0$) or short projects with poor fundamentals (which corresponds to $\underline{s} \leq 0$).

We still conjecture an equilibrium price function as (11) with $f(k)$ replaced by $f(x)$. We can show that the demand function and indirect utility function of transparent traders are still given by Equations (17) and (16), respectively, where $f_{\min}$ and $f_{\max}$ in (15) are amended to $f_{\min} \equiv \min_x f(x)$ and $f_{\max} \equiv \max_x f(x)$ accordingly to reflect the two-dimensional ambiguity feature. The market-clearing condition is still Equation (18). The new result in this extended economy is that two branches of Equation (17) are possible in equilibrium; that is, in equilibrium, transparent traders can hold either zero or a positive amount of stocks. This occurs because the extra dimension of ambiguity can make them so pessimistic about stock fundamentals that they completely withdraw from the market. We still compute the REE by considering the three possible values of Equation (17) case by case, which yields the following proposition (with proof provided in Appendix B):

Proposition A1. Suppose $0 < \mu < 1$. There exists an REE in the economy where transparent traders are ambiguous about multiple parameters of the structure of opaque traders' investment opportunities with price function

$$\bar{p} = \bar{v} + \bar{\theta}_T + \bar{\theta}_O - f(k, s),$$

where

$$f(k, s) = \sigma_\varepsilon s \sqrt{1 - \frac{1}{k}} + \frac{-(1 - \mu) \max\{f_{\min}, 0\} + \sigma_\varepsilon^2}{\mu k}$$

and

$$f_{\min} = \begin{cases} \frac{\sigma_\varepsilon \mu \underline{s} \sqrt{\bar{k}(\bar{k}-1)} + \sigma_\varepsilon^2}{1 - \mu + \mu \bar{k}} > 0, & \text{if } \mu < \frac{\sigma_\varepsilon}{(-\underline{s}) \sqrt{\bar{k}(\bar{k}-1)}}, \\ \sigma_\varepsilon \underline{s} \sqrt{1 - \frac{1}{k}} + \frac{\sigma_\varepsilon^2}{\mu k} \leq 0, & \text{if } \mu \geq \frac{\sigma_\varepsilon}{(-\underline{s}) \sqrt{\bar{k}(\bar{k}-1)}}. \end{cases}$$

In particular, if $\underline{s} = \bar{s} = 0$, Proposition A1 degenerates to Proposition 1 in our baseline model. We now explore the pricing implications of disclosure when the fraction μ of opaque traders is fixed. Let $\mathbf{x} \equiv (k, s)$ denote the true value of $x \equiv (k, s)$. As in the baseline model, we define the equity premium as $EP \equiv E(\bar{v} - \bar{p}) = f(\mathbf{x})$; that is, it is the average expected difference between the stock cash flow and the stock price, where the expectation is taken under the true value $\mathbf{x}$.

As we explained in our baseline model, requiring hedge funds to disclose their positions more frequently can be interpreted as a reduction in the degree of transparent traders' ambiguity. To capture this idea in this extended economy, we use $\gamma > 0$ to denote the disclosure level—that is, a higher γ means more disclosure—and assume that the bounds $(\underline{k}, \bar{k}, \underline{s}, \bar{s})$ are affected by γ in such a way that increasing γ will shrink the sizes of the ambiguity sets. Specifically, we define functions $\underline{k} = \underline{k}(\gamma)$, $\bar{k} = \bar{k}(\gamma)$, $\underline{s} = \underline{s}(\gamma)$, and $\bar{s} = \bar{s}(\gamma)$, and assume that $\underline{k}'(\gamma) \geq 0$, $\bar{k}'(\gamma) \leq 0$, $\underline{s}'(\gamma) \geq 0$, and $\bar{s}'(\gamma) \leq 0$. We can show that our Corollary 1 still holds; that is, reducing the amount of any type of ambiguity lowers the equity premium when μ is fixed. Formally we have the following corollary.

Corollary A1. When $\mu \in (0, 1)$ is fixed, increasing the disclosure level γ will (weakly) lower the equity premium EP in economies where transparent traders are ambiguous about multiple parameters of opaque traders' investment opportunities.

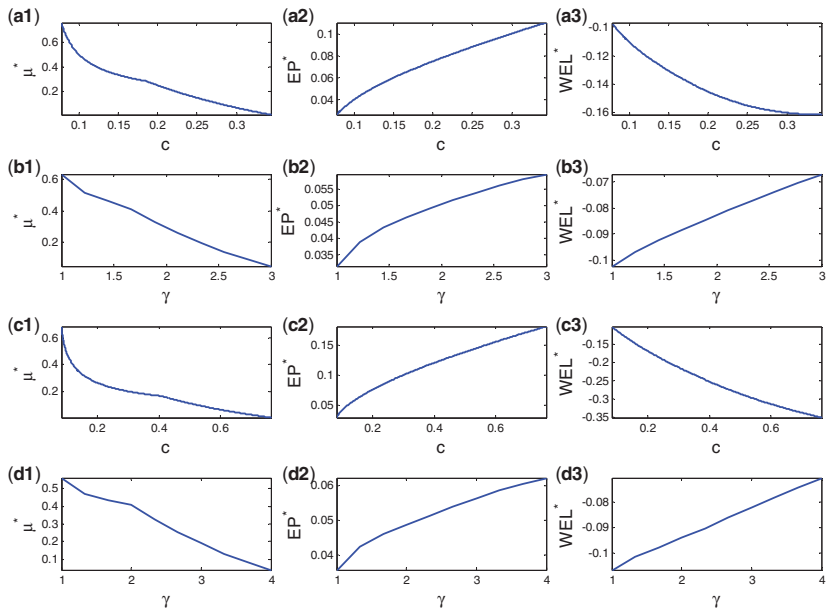


Figure A1
Implications of registration and disclosure in economies with multidimensional ambiguity about the private investment opportunities

In the top six panels (a1)–(b3), transparent traders are ambiguous about the Sharpe ratio s of opaque traders' extra investment opportunities, but not about their effective risk-tolerance parameter k . The bounds of its ambiguity set are determined by disclosure γ according to $\underline{s} = \frac{0.5}{\gamma}$ and $\bar{s} = \frac{0.5}{\gamma}$. In Panels (a1)–(a3), $\gamma=1$, and in Panels (b1)–(b3), $c=0.086$. In the bottom six panels (c1)–(d3), transparent traders are ambiguous about both s and k , and the bounds of the ambiguity sets are determined by disclosure γ according to $\underline{k} = 2 - \frac{1}{\gamma}$, $\bar{k} = 2 + \frac{1}{\gamma}$, $\underline{s} = -\frac{0.5}{\gamma}$, and $\bar{s} = -\frac{0.5}{\gamma}$. In Panels (c1)–(c3), $\gamma=1$, and in Panels (d1)–(d3), $c=0.091$. The other parameter values in all panels are: $\sigma_\varepsilon = 0.2$, $k = 2$, and $s = 0.5$.

In particular, if we set $\underline{s} = \bar{s} = 0$ so that $\underline{s}'(\gamma) = \bar{s}'(\gamma) = 0$, then Corollary A1 nests Corollary 1 in our baseline model as a special case. Alternatively, if we set $\underline{k}'(\gamma) = \bar{k}'(\gamma) = 0$, Corollary A1 still holds; that is, if disclosure serves only to reduce the ambiguity about the Sharpe ratio of opaque traders' extra investment opportunity, then more disclosure reduces the equity premium when the fraction μ of opaque traders is fixed.

In Appendix B, we further characterize the trader distribution equilibrium by computing the ex ante expected utilities of the two types of traders. The expressions of these utilities are complex. We therefore use a numerical example in Figure A1 to verify the robustness of Propositions 3 and 4. We still set $\sigma_\varepsilon = 0.2$ to match the empirical annual stock return volatility. We set the true values of the ambiguous parameters as $s=0$ and $k=2$. In the top six panels of Figure A1, we assume that transparent traders are ambiguous about the Sharpe ratio s of opaque traders' extra investment opportunities, but not about their effective risk-tolerance parameter k . The bounds of the ambiguity set are determined by the disclosure level γ according to the following forms: $\underline{s}(\gamma) = s - \frac{0.5}{\gamma}$ and $\bar{s}(\gamma) = s + \frac{0.5}{\gamma}$. In the bottom six panels of Figure A1, we allow transparent traders to be ambiguous about both s and k , and the bounds of the ambiguity sets are determined as: $\underline{s}(\gamma) = s - \frac{0.5}{\gamma}$, $\bar{s}(\gamma) = s + \frac{0.5}{\gamma}$, $\underline{k}(\gamma) = k - \frac{1}{\gamma}$, and $\bar{k}(\gamma) = k + \frac{1}{\gamma}$.

Comparing Figure A1 with Figures 1 and 2, we see that our Propositions 3 and 4 are robust in this extended economy. Specifically, in Panels (a1)–(a3) and (c1)–(c3) of Figure A1, increasing c

decreases the fraction μ^* of opaque traders, increases the equilibrium equity premium EP^* , and decreases the equilibrium welfare WEL^* , which is the same as Panels (b1)–(b3) of Figure 1 in our baseline model. Also, in Panels (b1)–(b3) and (d1)–(d3) of Figure A1, increasing disclosure level γ tends to decrease μ^* but increases EP^* and WEL^* . This is also consistent with Panels (b1)–(b3) of Figure 2, as an increase in γ in the extended economy corresponds to a decrease in Δk in our baseline model.

Multi-type opaque traders. In this subsection, we assume that the structures of the additional investment opportunities can be different among opaque traders, so that transparent traders will face multidimensional ambiguity. Specifically, opaque traders can access one of two types of investment opportunities $\tilde{\eta}_i$ that differ in their correlation $\rho_i \in (0, 1)$ with the randomness $\tilde{\varepsilon}$ of the stock payoff. Accordingly, we define $k_i \equiv \frac{1}{1-\rho_i^2}$ for $i = 1, 2$. Transparent traders are ambiguous about the two-dimensional parameter vector $k \equiv (k_1, k_2)$; that is, they view

$$k_i \in [k_i, \bar{k}_i] \text{ with } 1 \leq k_i \leq \bar{k}_i, \text{ for } i = 1, 2,$$

and cannot assign a probability on these sets. So again transparent traders face a two-dimensional ambiguity. We also follow the previous subsection and assume that the bounds of the ambiguity sets are related to disclosure $\gamma > 0$ by functions $\underline{k}_1 = k_1(\gamma)$, $\bar{k}_1 = \bar{k}_1(\gamma)$, $\underline{k}_2 = k_2(\gamma)$, and $\bar{k}_2 = \bar{k}_2(\gamma)$, where $\underline{k}'_1(\gamma) \geq 0$, $\bar{k}'_1(\gamma) \leq 0$, $\underline{k}'_2(\gamma) \geq 0$, and $\bar{k}'_2(\gamma) \leq 0$.

Information θ_T is still the public signal observable by all opaque and transparent traders, and information θ_O is the private signal observable only to opaque traders. We assume that once a trader becomes opaque, he has a probability of $\varphi_i \in [0, 1]$ of having access to $\tilde{\eta}_i$, where $\varphi_1 + \varphi_2 = 1$. Given a fraction μ of opaque traders active in the market, the mass of opaque traders who access $\tilde{\eta}_i$ is $\varphi_i \mu$. Our baseline model corresponds to the case of $\varphi_i = 1$ (for $i = 1, 2$).

The demand function $D_O(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k_i)$ and indirect utility $V_{O1}(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k_i)$ of opaque traders who access $\tilde{\eta}_i$ will still be given by Equations (8) and (29) with k replaced by k_i . Note that in this extension, even though those opaque traders accessing $\tilde{\eta}_i$ may be ambiguous about the trading strategies of other opaque traders accessing $\tilde{\eta}_j$ (i.e., ambiguous about parameter k_j), this ambiguity does not affect their trading decision problem. This is because all opaque traders have the same information $\{\tilde{\theta}_T, \tilde{\theta}_O\}$ and hence they do not infer information from prices (see also note 7 and the discussion at the beginning of Section 4).

The demand function $D_T(\tilde{p}, \tilde{\theta}_T)$ and indirect utility $V_{T1}(\tilde{p}, \tilde{\theta})$ will be given by Equations (17) and (31), respectively, except that $f_{\min}$ and $f_{\max}$ in (15) are now defined over a two-dimensional vector $k = (k_1, k_2)$. The market-clearing condition changes to

$$\varphi_1 \mu D_O(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k_1) + \varphi_2 \mu D_O(\tilde{p}, \tilde{\theta}_T, \tilde{\theta}_O; k_2) + (1 - \mu) D_T(\tilde{p}, \tilde{\theta}_T) = 1.$$

We then can follow similar steps as in our baseline model and show that there is an REE in this economy, which is characterized by the following proposition.

Proposition A2. Suppose $0 < \mu < 1$. There exists an REE in the economy with multi-type opaque traders, and the price function is

$$\tilde{p} = \bar{v} + \tilde{\theta}_T + \tilde{\theta}_O - f(k_1, k_2),$$

where

$$f(k_1, k_2) = \frac{\varphi_1 \bar{k}_1 + \varphi_2 \bar{k}_2}{\varphi_1 k_1 + \varphi_2 k_2} \frac{\sigma_{\tilde{\varepsilon}}^2}{(1 - \mu) + \mu (\varphi_1 \bar{k}_1 + \varphi_2 \bar{k}_2)}.$$

Clearly, if $\varphi_i = 1$ ($i = 1, 2$), then Proposition A2 degenerates to Proposition 1 in our baseline model. Let $\mathbf{k} \equiv (k_1, k_2)$ define the true values of k , and the equity premium is still defined as $EP = f(\mathbf{k})$. It is straightforward to show that reducing the amount of ambiguity about either parameter k_i will lower the equity premium EP . That is, Corollary 1 still holds. Formally, we have the following corollary.

Corollary A2. When $\mu \in (0, 1)$ is fixed, increasing the disclosure level γ will (weakly) lower the equity premium EP in the economy with multi-type opaque traders.

Now let us turn to the analysis of the trader distribution equilibrium. The ex ante expected utility V_{T0} of staying transparent is still given by (33), by noting that $f_{\min}$ and $f_{\max}$ in (15) are now defined over the two-dimensional vector (k_1, k_2) to reflect the fact of multidimensional ambiguity. However, the ex ante expected utility V_{O0} of becoming opaque is no longer generally given by (30). Specifically, given the recursive preference structure and the indirect utility $V_{O1}(\bar{p}, \bar{\theta}_T, \bar{\theta}_O; k_i)$ in (29) of opaque traders who access $\bar{\eta}_i$, we can compute:

$$V_{O0} = \min_{k_1 \in [\underline{k}_1, \bar{k}_1], k_2 \in [\underline{k}_2, \bar{k}_2]} E_k \left[\sum_i \varphi_i V_{O1}(\bar{p}, \bar{\theta}_T, \bar{\theta}_O; k_i) \right] \\ = -\exp \left(- \left[\bar{v} - c + \min_{(k_1, k_2)} \left[-f(k_1, k_2) - \log \left(\sum_i \varphi_i e^{-\frac{k_i}{2\sigma_\varepsilon^2} [f(k)]^2} \right) \right] - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right] \right).$$

The above equation reduces to (30) when $\varphi_i = 1$ (for $i = 1, 2$). The benefit of switching types is therefore given by

$$B(\mu) = \min_{(k_1, k_2)} \left[-f(k_1, k_2) - \log \left(\sum_i \varphi_i e^{-\frac{k_i}{2\sigma_\varepsilon^2} [f(k)]^2} \right) \right] - \left(-f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right).$$

Figure A2 uses a numerical example to illustrate the implications of registration (increasing c) and disclosure (increasing γ). We continue to set parameter σ_ε to 0.2 to reflect the empirical value

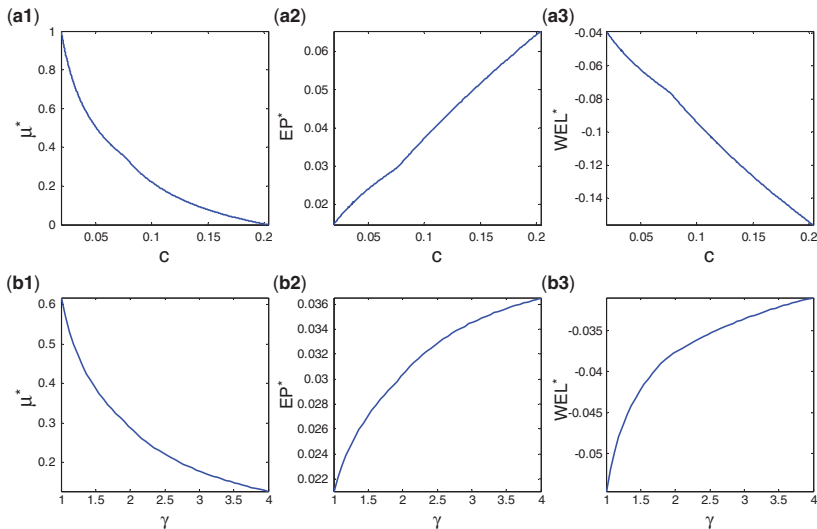


Figure A2
Implications of registration and disclosure in economies with multi-type opaque traders

This figure shows the implications of changing the cost c of switching trader types (in the top panels) and the degree γ of disclosure (in the bottom panels) for the equilibrium fraction μ^* of opaque traders, the equity premium EP^* , and welfare WEL^* , in the economies with two types of opaque traders. The two types of opaque-traders have different investment opportunities, and transparent traders are ambiguous about their structures. The bounds of the ambiguity sets are determined by disclosure γ according to $\underline{k}_1 = 2 - \frac{1}{\gamma}$, $\bar{k}_1 = 2 + \frac{1}{\gamma}$, $\underline{k}_2 = 3 - \frac{2}{\gamma}$, and $\bar{k}_2 = 3 + \frac{2}{\gamma}$. In Panels (a1)–(a3), $\gamma = 1$, and in Panels (b1)–(b3), $c = 0.038$. The other parameter values in all panels are: $\sigma_\varepsilon = 0.2$, $\varphi_1 = 0.3$, $\mathbf{k}_1 = 2$, and $\mathbf{k}_2 = 3$.

of the annual stock return volatility. We assume that the two types of investment opportunities are such that the true effective risk-tolerance parameters of opaque traders are equal to $\mathbf{k}_1=2$ and $\mathbf{k}_2=3$. Each opaque trader has a probability of $\varphi_1=0.3$ of having access to the first type of investment opportunity $\tilde{\eta}_1$. The bounds of the ambiguity sets are determined as: $\underline{k}_1(\gamma)=\mathbf{k}_1-\frac{1}{\gamma}$, $\bar{k}_1(\gamma)=\mathbf{k}_1+\frac{1}{\gamma}$, $\underline{k}_2(\gamma)=\mathbf{k}_2-\frac{2}{\gamma}$, and $\bar{k}_2(\gamma)=\mathbf{k}_2+\frac{2}{\gamma}$. We see that Propositions 3 and 4 continue to hold in this economy. In Panels (a1)–(a3) of Figure A2, increasing c decreases the fraction μ^* of opaque traders, increases the equilibrium equity premium EP^* , and decreases the equilibrium welfare WEL^* , which is the same as Panels (b1)–(b3) of Figure 1. In Panels (b1)–(b3) of Figure A2, increasing the disclosure level γ decreases μ^* but increases EP^* and WEL^* , and this is also consistent with Panels (b1)–(b3) of Figure 2.

Appendix B: Proofs and additional results

Compute the beta and alpha of opaque traders' portfolio. By Equation (23), Equation (26), and the definition of $\tilde{v}$ in (1),

$$\begin{aligned}\mathbf{Cov}(\tilde{R}_O, \tilde{R}_M | \tilde{p}) &= \mathbf{Cov}\left(\frac{1}{\tilde{p}-c} \left[\frac{(\tilde{\varepsilon} + f(k))f(k)}{(1-\rho^2)\sigma_\varepsilon^2} - \frac{\rho f(k)}{(1-\rho^2)\sigma_\varepsilon\sigma_\eta} \tilde{\eta} \right], \frac{\tilde{v}}{\tilde{p}} \middle| \tilde{p}\right) \\ &= \frac{f(\mathbf{k})}{(\tilde{p}-c)\tilde{p}} \mathbf{Cov}\left(\frac{\tilde{\varepsilon}}{(1-\rho^2)\sigma_\varepsilon^2} - \frac{\rho}{(1-\rho^2)\sigma_\varepsilon\sigma_\eta} \tilde{\eta}, \tilde{\varepsilon} \middle| \tilde{p}\right).\end{aligned}$$

Since the price $\tilde{p}$ in (11) only contains information about $\tilde{\theta}_T$ and $\tilde{\theta}_O$ and since $\mathbf{Cov}(\tilde{\eta}, \tilde{\varepsilon}) = \rho\sigma_\varepsilon\sigma_\eta$, we have

$$\mathbf{Cov}(\tilde{R}_O, \tilde{R}_M | \tilde{p}) = \frac{f(\mathbf{k})}{(\tilde{p}-c)\tilde{p}} \mathbf{Cov}\left(\frac{\tilde{\varepsilon}}{(1-\rho^2)\sigma_\varepsilon^2} - \frac{\rho}{(1-\rho^2)\sigma_\varepsilon\sigma_\eta} \tilde{\eta}, \tilde{\varepsilon}\right) = \frac{f(\mathbf{k})}{(\tilde{p}-c)\tilde{p}}.$$

By Equations (23), (1), and (11),

$$\mathbf{Var}(\tilde{R}_M | \tilde{p}) = \mathbf{Var}\left(\frac{\tilde{v}}{\tilde{p}} \middle| \tilde{p}\right) = \frac{1}{\tilde{p}^2} \mathbf{Var}(\tilde{\theta}_T + \tilde{\theta}_O + \tilde{\varepsilon} | \tilde{\theta}_T + \tilde{\theta}_O - f(\mathbf{k})) = \frac{\sigma_\varepsilon^2}{\tilde{p}^2}.$$

Thus,

$$\beta_O \equiv \frac{\mathbf{Cov}(\tilde{R}_O, \tilde{R}_M | \tilde{p})}{\mathbf{Var}(\tilde{R}_M | \tilde{p})} = \frac{\frac{f(\mathbf{k})}{(\tilde{p}-c)\tilde{p}}}{\frac{\sigma_\varepsilon^2}{\tilde{p}^2}} = \frac{\tilde{p} f(\mathbf{k})}{(\tilde{p}-c)\sigma_\varepsilon^2} = \frac{\tilde{p}}{\tilde{p}-c} \frac{EP}{\sigma_\varepsilon^2}$$

where the last equality follows from $EP = f(\mathbf{k})$. This proves Equation (27).

By Equations (26) and (11),

$$\begin{aligned}\mathbf{E}(\tilde{R}_O | \tilde{p}) &= \mathbf{E}\left(\frac{1}{\tilde{p}-c} \left[\frac{(\tilde{\varepsilon} + f(k))f(k)}{(1-\rho^2)\sigma_\varepsilon^2} - \frac{\rho f(k)}{(1-\rho^2)\sigma_\varepsilon\sigma_\eta} \tilde{\eta} \right] \middle| \tilde{\theta}_T + \tilde{\theta}_O - f(\mathbf{k})\right) \\ &= \frac{1}{\tilde{p}-c} \frac{[f(\mathbf{k})]^2}{(1-\rho^2)\sigma_\varepsilon^2} = \frac{1}{\tilde{p}-c} \frac{\mathbf{k}[f(\mathbf{k})]^2}{\sigma_\varepsilon^2},\end{aligned}$$

where the last equality follows from the definition of k in Equation (7). Also, by Equations (23), (1), and (11),

$$\mathbf{E}(\tilde{R}_M | \tilde{p}) = \mathbf{E}\left(\frac{\tilde{v}}{\tilde{p}} - 1 \middle| \tilde{p}\right) = \mathbf{E}\left(\frac{\tilde{p} + \tilde{\varepsilon} + f(k)}{\tilde{p}} - 1 \middle| \tilde{p}\right) = \mathbf{E}\left(\frac{\tilde{\varepsilon} + f(k)}{\tilde{p}} \middle| \tilde{p}\right) = \mathbf{E}\left(\frac{f(k)}{\tilde{p}} \middle| \tilde{p}\right) = \frac{f(\mathbf{k})}{\tilde{p}}.$$

So, $\alpha_O \equiv \mathbf{E}(\tilde{R}_O | \tilde{p}) - \beta_O \mathbf{E}(\tilde{R}_M | \tilde{p}) = \frac{1}{\tilde{p}-c} \frac{\mathbf{k}[f(\mathbf{k})]^2}{\sigma_\varepsilon^2} - \frac{\tilde{p} f(\mathbf{k})}{(\tilde{p}-c)\sigma_\varepsilon^2} \frac{f(\mathbf{k})}{\tilde{p}} = \frac{\mathbf{k}-1}{\tilde{p}-c} \frac{EP^2}{\sigma_\varepsilon^2}$, which proves Equation (28).

Proof of Proposition 2. We first give a more explicit expression of function B , defined by Equation (35). By Equation (21), we can express k as a function of $f(k)$:

$$k = \frac{\bar{k}\sigma_\varepsilon^2}{f(k)(1-\mu+\mu\bar{k})}.$$

Using the above expression, the minimization problem in the definition of B is simply:

$$\min_k \left[-f(k) + k \frac{[f(k)]^2}{2\sigma_\varepsilon^2} \right] = \min_{f(k)} \left[\left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right) f(k) \right].$$

Depending on the sign of the coefficient $\left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right)$, the minimized value is achieved at either $f_{\min}$ or $f_{\max}$. It can be easily shown that

$$\left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right) > 0 \Leftrightarrow \mu < \frac{(\bar{k}/2) - 1}{\bar{k} - 1}.$$

Thus, if $\bar{k} < 2$, then

$$B(\mu) = B_1(\mu) \equiv \left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right) f_{\max} - \left(-f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right),$$

and otherwise, the benefit function B can be further expressed as a combination of two branches:

$$B(\mu) = \begin{cases} B_0(\mu) \equiv \left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right) f_{\min} - \left(-f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right), & \text{if } \mu < \frac{\bar{k}/2 - 1}{\bar{k} - 1}, \\ B_1(\mu) \equiv \left(-1 + \frac{\bar{k}}{2(1-\mu+\mu\bar{k})} \right) f_{\max} - \left(-f_{\max} + \frac{f_{\min}^2}{2\sigma_\varepsilon^2} \right), & \text{otherwise,} \end{cases} \quad (\text{B1})$$

where the subscripts 0 and 1 in functions B_0 and B_1 indicate that those are values of B when μ is close to 0 and 1, respectively.

By Equation (21), we know:

$$f_{\min} = \frac{\sigma_\varepsilon^2}{1-\mu+\mu\bar{k}} \quad \text{and} \quad f_{\max} = \frac{\bar{k}\sigma_\varepsilon^2}{\bar{k}(1-\mu+\mu\bar{k})}. \quad (\text{B2})$$

Inserting these expressions into functions B_0 and B_1 delivers:

$$B_0(\mu) = \left[\frac{\bar{k}-\bar{k}}{\bar{k}} + \frac{\bar{k}-1}{2(1-\mu+\mu\bar{k})} \right] \frac{\sigma_\varepsilon^2}{1-\mu+\mu\bar{k}}, \quad (\text{B3})$$

$$B_1(\mu) = \frac{1}{2(1-\mu+\mu\bar{k})} \left(\frac{\bar{k}^2}{\bar{k}} - 1 \right) \frac{\sigma_\varepsilon^2}{1-\mu+\mu\bar{k}}. \quad (\text{B4})$$

Since $B_0(\mu) = B_1(\mu)$ at $\mu = \frac{(\bar{k}/2)-1}{\bar{k}-1}$ and since both B_0 and B_1 are decreasing in μ , function $B(\mu)$ is continuous and decreasing in μ .

The values of $\underline{c}$ and $\bar{c}$ in Proposition 2 are defined by $\underline{c} = B(1)$ and $\bar{c} = B(0)$. Clearly, $\underline{c} = B(1) = B_1(\mu) = \left(\frac{\bar{k}^2}{\bar{k}} - 1 \right) \frac{\sigma_\varepsilon^2}{2\bar{k}^2}$. For the value of $\bar{c}$, there are two possible cases. First, if $\bar{k} < 2$, then function B_0 is irrelevant, and $B(\mu) = B_1(\mu)$ for all values of $\mu \in [0, 1)$. So, $\bar{c} = B_1(0) = \frac{\sigma_\varepsilon^2}{2} \left(\frac{\bar{k}^2}{\bar{k}} - 1 \right)$. Second, if $\bar{k} \geq 2$, then for μ close to 0, $B(\mu) = B_0(\mu)$, and thus, $\bar{c} = B_0(0) = \left(\frac{2\bar{k}}{\bar{k}} + \bar{k} - 3 \right) \frac{\sigma_\varepsilon^2}{2}$. In sum, $\bar{c} = \min\{B_1(0), B_0(0)\}$. Thus, if $\underline{c} < c < \bar{c}$, then there is a unique interior equilibrium fraction $\mu^* \in (0, 1)$, which is determined by $B(\mu^*) = c$.

Finally, we show that $(\bar{c} - \underline{c})$ increases with Δk . By the definitions of $\bar{c}$ and $\underline{c}$, we have

$$(\bar{c} - \underline{c}) = \min \left\{ \frac{\bar{k}^2}{\underline{k}} - 1 - \left(\frac{1}{\underline{k}} - \frac{1}{\bar{k}^2} \right), \frac{2\bar{k}}{\underline{k}} + \bar{k} - 3 - \left(\frac{1}{\underline{k}} - \frac{1}{\bar{k}^2} \right) \right\} \frac{\sigma_\varepsilon^2}{2},$$

which is a continuous function of Δk . Because

$$\begin{aligned} \frac{\partial \left[\frac{\bar{k}^2}{\underline{k}} - 1 - \left(\frac{1}{\underline{k}} - \frac{1}{\bar{k}^2} \right) \right]}{\partial \Delta k} &= \frac{\bar{k}^2 - 1}{\underline{k}^2} + 2 \left(\frac{\bar{k}}{\underline{k}} - \frac{1}{\bar{k}^3} \right) > 0, \\ \text{and } \frac{\partial \left[\frac{2\bar{k}}{\underline{k}} + \bar{k} - 3 - \left(\frac{1}{\underline{k}} - \frac{1}{\bar{k}^2} \right) \right]}{\partial \Delta k} &= \frac{2\bar{k} - 1}{\underline{k}^2} + 1 + 2 \left(\frac{1}{\underline{k}} - \frac{1}{\bar{k}^3} \right) > 0, \end{aligned}$$

we have $\frac{\partial(\bar{c} - \underline{c})}{\partial \Delta k} > 0$.

Proof of Proposition 4. We first show that decreasing Δk will reduce the equilibrium fraction μ^* of opaque traders, by demonstrating that a decrease in Δk will shift downward both functions B_0 and B_1 defining the benefit function B in Equation (B1).

For a given μ , direct computation shows

$$\frac{\partial B_0}{\partial \Delta k} = \left[\frac{\bar{k} + \underline{k}}{\underline{k}^2} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} - \frac{\mu(\bar{k} - \underline{k})}{\underline{k}} \frac{1}{(1 - \mu + \mu\bar{k})} \right] \frac{\sigma_\varepsilon^2}{(1 - \mu + \mu\bar{k})}.$$

By $\mu < 1$:

$$\begin{aligned} & \frac{\bar{k} + \underline{k}}{\underline{k}^2} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} - \frac{\mu(\bar{k} - \underline{k})}{\underline{k}} \frac{1}{(1 - \mu + \mu\bar{k})} \\ & > \frac{\bar{k} + \underline{k}}{\underline{k}^2} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} - \frac{(\bar{k} - \underline{k})}{\underline{k}} \frac{1}{\bar{k}} \\ & = \frac{\bar{k}^2 + \underline{k}^2}{\underline{k}^2 \bar{k}} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2}. \end{aligned}$$

As long as $\frac{\bar{k}^2 + \underline{k}^2}{\underline{k}^2 \bar{k}} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} > 0$, then $\frac{\partial B_0}{\partial \Delta k} > 0$. We can show that this is indeed true for values of $\mu < \frac{\bar{k}/2 - 1}{\bar{k} - 1}$, when B_0 is relevant.

Specifically, if $\frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} > 0$, then clearly, $\frac{\bar{k}^2 + \underline{k}^2}{\underline{k}^2 \bar{k}} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} > 0$. If $\frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} < 0$, then:

$$\begin{aligned} \mu < \frac{\bar{k}/2 - 1}{\bar{k} - 1} &\Rightarrow \frac{1}{(1 - \mu + \mu\bar{k})} < \frac{2}{\bar{k}} \text{ and } \mu(\bar{k} - 1) < \bar{k}/2 - 1 \\ &\Rightarrow \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} > \frac{1 - (\bar{k}/2 - 1)}{2} \left(\frac{2}{\bar{k}} \right)^2, \end{aligned}$$

and hence

$$\frac{\bar{k}^2 + \underline{k}^2}{\underline{k}^2 \bar{k}} + \frac{1 - \mu(\bar{k} - 1)}{2(1 - \mu + \mu\bar{k})^2} > \frac{\bar{k}^2 + \underline{k}^2}{\underline{k}^2 \bar{k}} + \frac{1 - \bar{k}/2 + 1}{2} \left(\frac{2}{\bar{k}} \right)^2 = \frac{\bar{k}^2 \bar{k} + 4\underline{k}^2}{\underline{k}^2 \bar{k}^2} > 0.$$

For a given value of μ , direct computation shows

$$\frac{\partial B_1}{\partial \Delta k} = \left[\left(2\bar{k}\underline{k} + \bar{k}^2 \right) - 2 \frac{\mu(\bar{k}^2 \underline{k} - \underline{k}^2)}{(1 - \mu + \mu\bar{k})} \right] \frac{1}{\underline{k}^2 (1 - \mu + \mu\bar{k})^2}.$$

Note that $\frac{\mu(\bar{k}^2 k - k^2)}{(1 - \mu + \mu \bar{k})}$ is increasing in μ , and thus

$$\left(2\bar{k}\underline{k} + \bar{k}^2\right) - 2\frac{\mu(\bar{k}^2 k - k^2)}{(1 - \mu + \mu \bar{k})} > \left(2\bar{k}\underline{k} + \bar{k}^2\right) - 2\frac{(\bar{k}^2 k - k^2)}{\bar{k}} = \bar{k}^2 + 2\frac{k^2}{\bar{k}} > 0.$$

So, $\frac{\partial B_1}{\partial \Delta k} > 0$.

We next show that decreasing Δk will reduce the equity premium when $\mu^* > \frac{\bar{k}/2 - 1}{\bar{k} - 1}$. When $\mu^* > \frac{\bar{k}/2 - 1}{\bar{k} - 1}$, function B_1 is relevant. Thus, the equilibrium condition for the trader-type decision is

$$B_1(\mu^*) = \frac{1}{2(1 - \mu^* + \mu^* \bar{k})} \left(\frac{\bar{k}^2}{\underline{k}} - 1 \right) \frac{\sigma_\varepsilon^2}{1 - \mu^* + \mu^* \bar{k}} = c, \quad (B5)$$

which implies

$$EP^* = \frac{\bar{k}\sigma_\varepsilon^2}{\mathbf{k}(1 - \mu^* + \mu^* \bar{k})} = \sqrt{\frac{1}{\underline{k}^{-1} - \bar{k}^{-2}}} \frac{\sqrt{2c\sigma_\varepsilon^2}}{\mathbf{k}}. \quad (B6)$$

Clearly, a decrease in Δk will increase EP^* .

In addition, Equation (B5) also implies an analytical expression of μ^* :

$$\mu^* = \frac{\sqrt{\frac{(\bar{k}^2 \underline{k}^{-1} - 1)\sigma_\varepsilon^2}{2c}} - 1}{\bar{k} - 1}. \quad (B7)$$

Putting this expression into the condition of $\mu^* > \frac{\bar{k}/2 - 1}{\bar{k} - 1}$ can characterize this condition in terms of exogenous parameters.

Plugging Equation (B6) into Equation (38), we can express the welfare as follows:

$$WEL^* = -\sqrt{\frac{2c\sigma_\varepsilon^2}{\underline{k} - (\underline{k}/\bar{k})^2}} + \frac{c}{\bar{k}^2 \underline{k}^{-1} - 1}, \quad (B8)$$

where the first term captures the expected wealth and the second term captures the trading benefit. Clearly, decreasing Δk will increase the second term. Direct computation shows that

$$\frac{\partial}{\partial \Delta k} \left(-\sqrt{\frac{2c\sigma_\varepsilon^2}{\underline{k} - (\underline{k}/\bar{k})^2}} \right) > 0 \Leftrightarrow 2\left(\frac{\underline{k}\bar{k} + \bar{k}^2}{\bar{k}^3}\right) < \bar{k}^3 \Leftrightarrow \underline{k} < \frac{\bar{k}}{2}(\sqrt{1 + 2\bar{k}} - 1).$$

Thus, if $\underline{k} < \frac{\bar{k}}{2}(\sqrt{1 + 2\bar{k}} - 1)$, then $\frac{\partial WEL^*}{\partial \Delta k} > 0$.

Computing the ex ante expected utilities of traders in Section 4. The indirect utility of being a transparent trader in a market with price p^{sym} is:

$$V_{T1}^{sym}(\tilde{p}^{sym}) = -e^{-\left\{ \tilde{p}^{sym} + \frac{[f^{sym}(k)]^2}{2\sigma_\varepsilon^2} \right\}} = -e^{-\left\{ \bar{v} + \bar{\theta}_T - f^{sym}(k) + \frac{[f^{sym}(k)]^2}{2\sigma_\varepsilon^2} \right\}}.$$

Thus, by the recursive multiple-priors structure, we have

$$V_{T0}^{sym} = \min_k V_{T1}^{sym}(\tilde{p}^{sym}) = -\exp \left[\bar{v} + \min_k \left\{ -f^{sym}(k) + \frac{[f^{sym}(k)]^2}{2\sigma_\varepsilon^2} \right\} - \frac{\sigma_{\theta T}^2}{2} \right]. \quad (B9)$$

Note that the quadratic function $y = -x + \frac{x^2}{2\sigma_\varepsilon^2}$ is downward sloping at the range of $(0, \sigma_\varepsilon^2)$ and that $f^{sym}(k) \in (0, \sigma_\varepsilon^2)$ by Equation (42). Thus,

$$\min_k \left\{ -f^{sym}(k) + \frac{1}{2} \frac{[f^{sym}(k)]^2}{\sigma_\varepsilon^2} \right\} = -f_{\max}^{sym} + \frac{1}{2} \frac{(f_{\max}^{sym})^2}{\sigma_\varepsilon^2}.$$

Plugging the above equation into Equation (B9) delivers Equation (44).

The indirect utility of being an opaque trader in a market with price $\bar{p}^{sym}$ is:

$$\begin{aligned} V_{O1}^{sym}(\bar{p}^{sym}) &= \exp \left[- \left(\bar{p}^{sym} - c + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 \right) \right] \\ &= \exp \left[- \left(\bar{v} - c - f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 - \frac{\sigma_{\theta T}^2}{2} \right) \right]. \end{aligned}$$

So, the ex ante expected utility of becoming opaque is:

$$V_{O0}^{sym} = \min_k V_{O1}(\bar{p}^{sym}) = \exp \left[- \left(\bar{v} - c + \min_k \left[-f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 \right] - \frac{\sigma_{\theta T}^2}{2} \right) \right]. \quad (B10)$$

By Equation (42), we have

$$k = \frac{\sigma_\varepsilon^2}{\mu f^{sym}(k)} - \frac{1-\mu}{\mu}.$$

Inserting the above equation into $-f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2$ yields:

$$-f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 = -\frac{2\mu-1}{2\mu} f^{sym}(k) - \frac{1-\mu}{2\sigma_\varepsilon^2 \mu} [f^{sym}(k)]^2.$$

So, if $\frac{2\mu-1}{2\mu} > 0$, that is, if $\mu > \frac{1}{2}$, the quadratic function $y = -\frac{2\mu-1}{2\mu}x - \frac{1-\mu}{2\sigma_\varepsilon^2 \mu}x^2$ is decreasing in the range of $(0, \infty)$, so that the minimum of $\left[-f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 \right]$ is achieved at $f_{\max}^{sym}$ (when k is equal to $\underline{k}$). That is, if $\mu > \frac{1}{2}$, we have:

$$\min_k \left[-f^{sym}(k) + \frac{k}{2\sigma_\varepsilon^2} [f^{sym}(k)]^2 \right] = -\frac{2\mu-1}{2\mu} f_{\max}^{sym} - \frac{1-\mu}{2\sigma_\varepsilon^2 \mu} (f_{\max}^{sym})^2 = -f_{\max}^{sym} + \frac{k}{2\sigma_\varepsilon^2} (f_{\max}^{sym})^2.$$

Plugging the above equation into Equation (B10) delivers Equation (46).

Proof of Proposition A1. We discuss market-clearing case by case. First, suppose $D_T(\bar{p}, \bar{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$. Substituting this expression into the market-clearing condition, solving for the equilibrium price and comparing with the conjectured price function, we have:

$$f(x) = \sigma_\varepsilon \sqrt{1 - \frac{1}{k} + \frac{-(1-\mu)f_{\min} + \sigma_\varepsilon^2}{\mu k}}. \quad (B11)$$

In the current case of $D_T(\bar{p}, \bar{\theta}_T) = \frac{f_{\min}}{\sigma_\varepsilon^2}$ we must have $f_{\min} > 0$, and hence $f(x) > 0$ for all possible x . In particular, $f(k, \underline{s}) = \sigma_\varepsilon \sqrt{1 - \frac{1}{k} + \frac{-(1-\mu)f_{\min} + \sigma_\varepsilon^2}{\mu k}} > 0$, which in turn implies $-(1-\mu)f_{\min} + \sigma_\varepsilon^2 > 0$. As a result, $f(x)$ achieves its minimum at $x = (\bar{k}, \underline{s})$:

$$f_{\min} = \sigma_\varepsilon \sqrt{1 - \frac{1}{\bar{k}} + \frac{-(1-\mu)f_{\min} + \sigma_\varepsilon^2}{\mu \bar{k}}} \Rightarrow f_{\min} = \frac{\sigma_\varepsilon \mu \sqrt{\bar{k}(\bar{k}-1)} + \sigma_\varepsilon^2}{1 - \mu + \mu \bar{k}}. \quad (B12)$$

In particular, when $\underline{s} = 0$, we have Equation (20) as in our baseline model. Plugging (B12) into (B11) yields the explicit expression of $f(x)$ in this case. In the current case, we also we need to impose the following condition:

$$f_{\min} > 0 \Rightarrow \mu < \frac{\sigma_\varepsilon}{(-\underline{s})\sqrt{\bar{k}(\bar{k}-1)}}; \quad (B13)$$

that is, (B11) is an equilibrium when the fraction μ of opaque traders is small.

Second, suppose that $D_T(\tilde{p}, \tilde{\theta}_T) = 0$. Then, we can follow the same procedure as in the first case and compute:

$$f(x) = \sigma_\varepsilon s \sqrt{1 - \frac{1}{k} + \frac{\sigma_\varepsilon^2}{\mu k}}. \quad (\text{B14})$$

Again, we need to check the condition of $f_{\min} \leq 0 \leq f_{\max}$ to ensure that the above equation forms an equilibrium. Clearly, $f_{\max} \geq 0$ is always satisfied, since $f_{\max} \geq f(k, \bar{s}) > 0$. In the current case, $f(x)$ still achieves its minimum at $x = (\bar{k}, \bar{s})$:

$$f_{\min} = \sigma_\varepsilon \bar{s} \sqrt{1 - \frac{1}{\bar{k}} + \frac{\sigma_\varepsilon^2}{\mu \bar{k}}}. \quad (\text{B15})$$

So,

$$f_{\min} \leq 0 \Rightarrow \mu \geq \frac{\sigma_\varepsilon}{(-\bar{s})\sqrt{\bar{k}(\bar{k}-1)}}; \quad (\text{B16})$$

that is, (B14) is an equilibrium when μ takes large values.

Last, suppose $D_T(\tilde{p}, \tilde{\theta}_T) = \frac{f_{\max}}{\sigma_\varepsilon^2}$. Then, we can show that $f(x) = \sigma_\varepsilon s \sqrt{1 - \frac{1}{k} + \frac{-(1-\mu)f_{\max} + \sigma_\varepsilon^2}{\mu k}}$. However, this case requires $f_{\max} < 0$, which in turn means $-(1-\mu)f_{\max} + \sigma_\varepsilon^2 > 0$ and hence $f_{\max} \geq \bar{s} \sqrt{1 - \frac{1}{\bar{k}}} \sigma_\varepsilon + \frac{-(1-\mu)f_{\max} + \sigma_\varepsilon^2}{\mu \bar{k}} > 0$. This is a contradiction. Therefore, this case is off equilibrium. Summarizing the above three cases delivers Proposition A1.

Characterization of trader distribution equilibrium in Appendix A. The proof of Proposition A1 shows that transparent traders' demand function is

$$D_T(\tilde{p}; \tilde{\theta}_T) = \max \left\{ 0, \frac{f_{\min}}{\sigma_\varepsilon^2} \right\}. \quad (\text{B17})$$

So, plugging Equation (B17) into Equation (16), we compute the indirect utility of a transparent trader:

$$V_{T1}(\tilde{p}, \tilde{\theta}_T) = \min_x E_x \left[-e^{-(\tilde{p} + D_T(\tilde{p}, \tilde{\theta}_T)(\bar{v} - \tilde{p}))} \right] = -\exp \left[-\left(\tilde{p} + \frac{(\max\{0, f_{\min}\})^2}{2\sigma_\varepsilon^2} \right) \right]. \quad (\text{B18})$$

Given the recursive multiple-priors utility representation, the ex ante utility of staying transparent is:

$$\begin{aligned} V_{T0} &= \min_x E_x [V_{T1}(\tilde{p}, \tilde{\theta}_T)] \\ &= -\exp \left[-\left(\bar{v} - f_{\max} + \frac{(\max\{0, f_{\min}\})^2}{2\sigma_\varepsilon^2} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \right) \right], \end{aligned} \quad (\text{B19})$$

where the second equation follows from Equations (11) and (B18), and where

$$f_{\max} \equiv \max_x f(x) = \max_k \left[\sigma_\varepsilon \bar{s} \sqrt{1 - \frac{1}{k} + \frac{-(1-\mu)\max\{f_{\min}, 0\} + \sigma_\varepsilon^2}{\mu k}} \right], \quad (\text{B20})$$

by the expression of $f(x)$ in Proposition A1.

Applying a similar analysis to opaque traders with the help of Equations (11) and (A5), we can compute the ex ante utility of becoming opaque as follows:

$$\begin{aligned} V_{O0} &= \min_x E_x [V_{O1}(\tilde{p}, \tilde{\theta}_O; x)] \\ &= -\exp \left[-\left(\begin{aligned} &\bar{v} - c - \frac{\sigma_{\theta T}^2 + \sigma_{\theta O}^2}{2} \\ &+ \min_x \left[-f(x) + k \frac{[f(x)]^2}{2\sigma_\varepsilon^2} - \frac{s\sqrt{k(k-1)}}{\sigma_\varepsilon} f(x) + \frac{ks^2}{2} \right] \end{aligned} \right) \right]. \end{aligned} \quad (\text{B21})$$

Thus, the benefit of switching from being a transparent trader to an opaque trader is:

$$B(\mu) = \min_x \left[-f(x) + k \frac{[f(x)]^2}{2\sigma_\varepsilon^2} - \frac{s\sqrt{k(k-1)}}{\sigma_\varepsilon} f(x) + \frac{ks^2}{2} \right] - \left(-f_{\max} + \frac{(\max\{0, f_{\min}\})^2}{2\sigma_\varepsilon^2} \right). \quad (\text{B22})$$

Still, the equilibrium fraction μ^* is determined by setting $B(\mu^*) = c$, and welfare can be measured by the certainty equivalent of a transparent trader's ex ante equilibrium utility (adjusted by subtracting a constant $\bar{v} - \frac{\sigma_{\theta T}^2 + \sigma_{\theta Q}^2}{2}$):

$$WEL^* = -f_{\max}^* + \frac{(\max\{0, f_{\min}^*\})^2}{2\sigma_\varepsilon^2}. \quad (\text{B23})$$

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